

Minority Salience and Violent Crime

Job Market Paper

Grigory Aleksin*

October 7, 2026

Abstract

This paper investigates how changes in the salience of Muslim communities impact local violent crime rates in England and Wales. Temporal shocks to Muslim salience are driven by terrorist attacks attributed to Muslim perpetrators. Leveraging police recorded crime data at the neighbourhood level, I use a matched difference-in-difference approach that exploits variation in the distribution of Muslim communities and the presence of mosques across England and Wales. The results show that, following a terrorist attack, violent crime increases by up to 13% in “Muslim” neighbourhoods, persisting for about six months. Moreover, I find evidence that the spike in violence is spatially diffuse and spills over to other minority groups. The impact is also larger in areas with greater religious fractionalization and higher support for right-wing parties. More broadly, the salience-crime link holds for other minority groups including Jewish communities, following the intensification of the Israeli-Palestinian conflict.

Keywords: violent crime, terrorism, minorities, discrimination

JEL Codes: D91, J15, K42, Z12, Z13

*Royal Holloway University of London. Email: grigory.aleksin@rhul.ac.uk

1 Introduction

The last two decades have seen substantial growth in ethnic and religious diversity across many Western economies.¹ This major societal change has coincided with reduced economic growth, a deterioration of government fiscal balance, and a seemingly continuous series of international crises (e.g. 2008/09 Financial Crisis, 2015 European Migrant Crisis, Brexit, COVID-19, Ukraine War, Conflicts in Israel and the Middle East). The subsequent growth of right-wing populism, expressing opposition to migrants and other minorities, has prompted intense debate about whether this emerging sentiment has economic or socio-cultural roots, and what governments can do to instead promote trust and social cohesion (Alesina and Tabellini, 2024; Guriev and Papaioannou, 2022). The UK is no exception. Economic growth has been low for decades, it has enacted significant austerity policies and left the European Union partly on the promise of greater control over its borders (Becker and Fetzer, 2017; Fetzer, 2019). At the same time, the extreme right party has become one of the most popular, and the number of hate crimes has grown year-on-year.²

In this paper, I contribute to the understanding of the growing hostility towards minorities. I focus on changes in the salience of minority groups, whereby the attention of the public is disproportionately drawn to concerns about these communities. Salience can magnify existing prejudices and expose their sources, whether economic or cultural (Bonomi et al., 2021; Bordalo et al., 2013). For instance, during Ramadan, electoral support for far-right or nationalist parties in Germany increases in areas closer to Mosques, where locals become more aware of their Muslim neighbours (Colussi et al., 2021). I focus on a more extreme and negative source of shocks to salience: Islamic terrorist attacks. In a well-integrated society such events should not be attributed or blamed on local Muslim communities, and no backlash should be observed (Shayo, 2020). Sadly, this is not what I find.

I show a large increase in violent crime following an Islamic terrorist attack. Using a matched difference-in-differences design that exploits the locations of mosques and large Muslim communities, I find that an Islamist attack raises violent crime in exposed neighbourhoods by nearly 13 percent, and the effect persists for up to six months. Attacks differ widely in how much attention they receive. To capture this, I also measure the shock directly by the intensity of newspaper coverage and Google search activity. These measures give consistent results. Violent crime rises by 5.5 and 4.5 percent following surges in coverage and search activity, and by 12 and 9 percent in the months of peak attention. The backlash also spills over to nearby neighbourhoods with large non-Muslim ethnic minority populations. The results hold under a Bartik-style design and a battery of further robustness checks.

I conduct a detailed heterogeneity analysis comparing the magnitude of effects across a number of socio-economic characteristics as well as the fall in governmental funding driven by the fiscal austerity measures. These speak directly to whether hostility towards minorities and immigrants is rooted in neighbourhood and constituency-level characteristics related to deprivation/financial fragility (Fetzer, 2019; Liberini et al., 2019), or social factors such as a lack of contact (Alba et al., 2005; Alesina et al., 2023; Alrababa’H et al., 2021). I find little evidence that socio-economic factors affect these results. Instead effects are largest in neighbourhoods with the greatest religious frac-

¹From 2000 to 2020 the share of foreign-born residents in OECD countries rose from 8.9% to 12.2%.

²For more information see: <https://www.gov.uk/government/statistics/hate-crime-england-and-wales-year-ending-march-2025/hate-crime-england-and-wales-year-ending-march-2025>.

tionalization and the lowest shares of young people aged 0-24. In addition, I find that political preferences, proxied by voting behaviour, appear to matter as well. Prior work has explored how political preferences can reveal private sentiments, and lead to a deterioration of social norms (Bursztyn et al., 2020; Clifton-Sprigg et al., 2020). I use data from the 2010 and 2015 General Elections, and the 2016 Brexit Vote to check whether the backlash in areas that voted for right-wing anti-immigrant parties (UKIP/BNP) or for ‘Leave’ in the 2016 United Kingdom European Union membership (Brexit) referendum, saw larger crime responses to changes in salience. Indeed, I find gaps in effects between areas with high vs. low support for xenophobic parties. However, I go further and consider whether the heterogeneity in these effects is explained by differences in the baseline socio-economic characteristics. To do so, following Illing et al. (2024), I use a re-weighting strategy so that the socio-economic characteristics of low-voting areas match those of high-voting ones. This approach reveals that socio-demographic characteristics, such as those given above, are not sufficient to explain most of the gap in the effects. These results suggest that hostility towards Muslims, and by extension other minority groups, can be partly attributed to prejudices that arise independently of local socio-economic conditions, perhaps exacerbated by traditional and social media platforms (Bursztyn et al., 2019; DeMarzo et al., 2003; Schneider-Strawczynski and Valette, 2025).

This paper relates to several strands of the existing literature. It builds on recent empirical work evaluating the consequences of terrorism on the integration of Muslim communities, voting outcomes, political attitudes or hate crimes (Böhmelt et al., 2020; Deloughery et al., 2012; Gould and Klor, 2016; Hanes and Machin, 2014; Ivandic et al., 2024; Swahn et al., 2003). These find that terrorist attacks deter integration, and strengthen existing hostility from locals, leading to more widespread support for far-right and nationalist parties, and even retaliatory hate crimes. Closest to this paper is Ivandic et al. (2024) that estimates the short-term impact of ten different Islamic terrorist attacks on hate crimes committed in the Greater Manchester area by comparing crime trends across several minority groups. This paper, on the other hand, uses police recorded data from neighbourhoods throughout England and Wales. This not only allows for greater flexibility when it comes to identification, but also provides richer cross-sectional variation for heterogeneity analysis.

More generally, there is also a newly emerging literature documenting the importance of political shocks, such as the election of Donald Trump or the decision of the UK to leave the EU, and the consumption of social media content as determinants of hate crime (Clifton-Sprigg et al., 2020; Müller and Schwarz, 2021, 2023; Rushin and Edwards, 2018). These studies highlight how such electoral results, that reveal the extent of anti-minority sentiment, or “echo chambers” that reinforce existing beliefs can rapidly erode previously held social norms. However, unlike this paper, these studies do not consider the socio-economic conditions that make the emergence of hate crime more or less likely.

Additionally, I contribute to the literature debating the underlying causes of growing anti-minority and anti-immigrant sentiment (Alesina and Tabellini, 2024). Situated within this are attempts to analyse the determinants of the *Leave* vote in the Brexit referendum, which was driven primarily by concerns about immigration (Alabrese et al., 2019; Becker et al., 2017). It remains unclear whether these concerns were rooted in the worsening economic circumstances or simply misperceptions about migrant communities and their impact on wider society. As in Alesina et al. (2023) this paper provides evidence in favour of the latter view that social factors, that make individuals prone to misinformation, play a more important explanatory role.

The rest of the paper is organized as follows. Section 2 describes the data for crime, the distribution of Muslim communities around the UK, Muslim salience and the control variables. In Section 3, I discuss my identification strategy and robustness checks. Section 4 presents the results, and finally section 5 concludes the paper.

2 Data

Before examining the data, it is useful to first define the geographic units used in the analysis. Most of the data are provided in terms of three geographies: Lower Layer Super Output Areas (LSOAs), Westminster Parliamentary Constituencies and Local Authority Districts (LADs).³ LSOAs are the smallest, with around 1500 residents, and will be referred to as neighbourhoods. Westminster Parliamentary Constituencies (often referred to as simply *Constituencies*) are large units with just over 100,000 residents that are used to elect local Members of Parliament (MPs). As one focus of our analysis concerns political preferences, I will primarily work with constituencies when performing heterogeneity analysis. Lastly, LADs (often called districts) are the main administrative units in the UK. There are approximately 350 such units that are responsible for local government policies and public good provision. Since LADs are the most recognisable by most residents, these will be used in my treatment definitions. Note that, except for the analysis of the Israeli-Palestinian Conflict, the 2011 full extent boundaries are used for the aforementioned geographic units. Note that LSOAs are nested within LADs, this is not the case for constituencies.

2.1 Muslim Communities

Information about the religious composition of residents is taken from the 2011 Census, and measured as the share of residents within an LSOA that identify as Muslim. Although Islam is still a minority religion in the UK, the number of adherents has risen considerably over the last two decades, with the share of Muslims in the general population rising from 3% (2001) to 4.8% (2011) to 6.5% (2021), and the number of mosques (or masjids) growing commensurately. Most of this growth has clustered around historical Muslim communities that provide support networks and aid with employment. These are former industrial cities, such as Birmingham, Leeds and Sheffield, where many Muslim migrants settled in the post Second World War economic expansion. In recent decades, as structural change reduced the share of manufacturing in the UK economy, these cities have experienced poorer economic performance, sparking tensions between different ethnic and religious groups, sometimes resulting in violent hate crimes.

Mosques often became the target or setting for such violent encounters.⁴ Not only are they a visible marker for the presence of Muslims in the area, but they are a regular gathering place for the religious community, leading to them becoming targets of anti-Muslim actions. Both mosque locations and neighbourhood religious affiliation are used to define ‘treated’ areas. Mosque locations are provided by <https://www.muslimsinbritain.org/index.php> which hosts a definitive directory of all Muslim places of worship around the UK. In particular, I use the December 2019 edition of the data, which contains 2131 records countrywide.⁵ To make sure that only ‘visi-

³Whenever another is used, it will be noted and explained in a separate footnote.

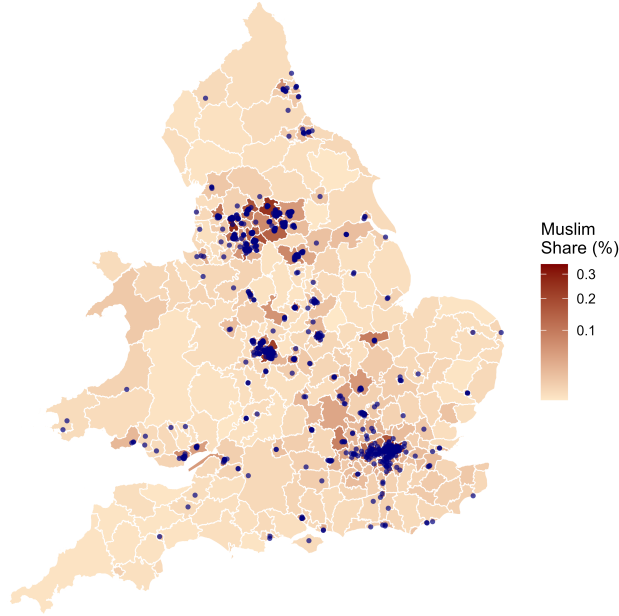
⁴For an in-depth discussion see the report by the British Muslim Trust available at https://britishmuslimtrust.co.uk/assets/reports/A_Summer_of_Division_BMT_Report.pdf.

⁵Past versions are not used as they had significant omissions and would introduce large measurement error into the analysis. The disadvantage of this approach is that I am not able to observe the

ble' places of worship are considered, I focus only on entries referred to as mosques or masjids. Hence, I exclude prayer rooms found in hospitals, student union buildings, or airports, as well as multi-faith rooms/centres.

Most mosques are located in older community centres or buildings previously used by the Church of England (Church or parish buildings) that were sold to local Muslim communities. This means that the creation of a new Mosque is almost entirely demand-driven. Therefore, it is natural to use the presence of a Mosque as a visible marker of the presence of a relatively large Muslim community in the area. Figure 1 plots the country-wide distribution of Mosques, alongside a heatmap for the share of self-identified Muslims at the district-level. Unsurprisingly, the two are strongly correlated, and, as noted above, are largest in urban areas.

Figure 1: Plot of Mosque Locations and Muslim Share



Notes: A plot of the locations of mosques (blue dots) across England and Wales, alongside a heatmap for the percentage share of Muslim residents at the local authority district-level.

For the analysis, I use the distribution of Mosques and Muslim residents to create a single measure of exposure to changes in salience. In particular, I define a “treated” or “exposed” neighbourhood as one that is in the 75th percentile for the share of Muslim residents in its local authority district, and is within 500m of a Mosque. The combined measure makes sure that a neighbourhood has both a visible and relatively large community of Muslim residents. Put in economic terms, these conditions imply lower search costs for violent criminals, that are responding to a change in salience. Table 1 is a contingency table summarising the treatment variable. About 6% of all neighbourhoods are within 500m of a mosque, and two-thirds of those have a relatively large share of

construction of new Mosques over the sample period. However, if anything, this would lead to an under-estimation of results because I would consider as treated those neighbourhoods that do not yet contain a mosque.

Muslim residents (at the 75th percentile). This yields a total of just under 1500 ‘treated’ neighbourhoods. To make sure that the results are not driven by this specific choice, I show that they are robust to several other definitions of exposure.

Table 1: Contingency Table of Mosques and Muslim Share (%)

	Not Top Quartile	Top Quartile
No Mosque	72.59	21.36
Mosque	1.99	4.05

Notes: This contingency contains the percentage of all neighbourhoods split by whether they are within 500m of a mosque, or whether the share of residents that identify as Muslim in that neighbourhood is in the top quartile of the local authority district.

2.2 Muslim Salience

My baseline results will be presented for three related measures of salience. The first, and most straightforward, is a binary indicator for whether a major terrorist attack attributed to a Muslim group has occurred in a given month based on the data available in the Global Terrorism Database. To avoid using attacks that would not be noticed by UK residents, I make several sample restrictions. Specifically, I require that any terrorist attacks, attributed to Islamic groups, need to have taken place in the UK, Western Europe or the US, and, for the latter two, to have at least 5 fatalities. As a result, my sample contains 33 Islamic terrorist attacks, occurring mostly in the 2015-2017 period. This means that my focus is primarily on those attacks where the victims are those living in the ‘West’.⁶

The second measure is based on the number of news articles, from UK-based sources, written about Islamic terrorism each month, as recorded by LexisNexis.⁷ This includes articles both online and in-print, in national and local newspapers. To make sure that the articles are relevant, I use the following rule as a filter: “(terror or terrorist or terrorism) & (attack or bomb or bombing or incident) & (muslim or islam or islamist or jihadi or isil or al-qaeda or isis or islamic state)” (Ivandic et al., 2024). The number of articles is highly responsive to the occurrence of a terrorist attack: on average, in months where such an attack has occurred, the number of articles written jumps from 933 to 2472. To make this metric comparable to the one below, in the analysis, I use a normalised measure where the month with the highest number of published articles is assigned the value 100, and all others are expressed relative to it.

The third measure is based on Google Trends data for the UK at the monthly-level. This is a metric of search interest, that Google normalises into a 0-100 index relative to the month with the greatest search activity. Specifically, I use the values for the search terms: “islamic terrorist”.⁸ Google search activity reflects the same pattern as newspaper articles: the average value of the index is about three times higher in months when an attack occurs.

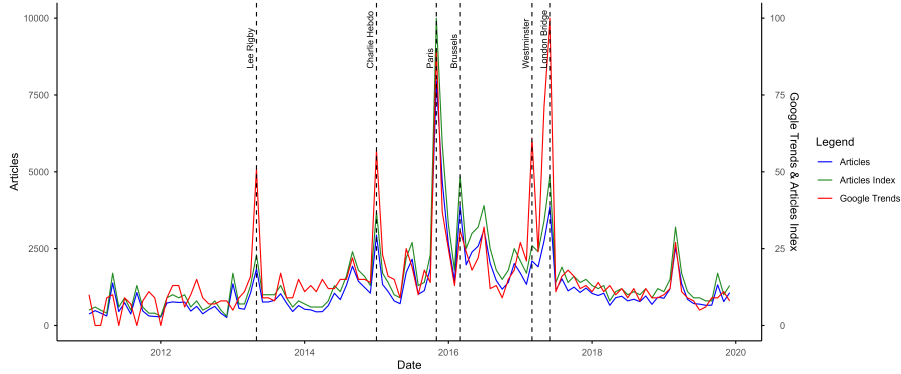
⁶The criteria are similar to those used in Ivandic et al. (2024). However, I find a larger number of attacks in my sample period.

⁷LexisNexis is a data repository for legal information, news/media articles, public records, financial information and other data. For more information see <https://www.lexisnexis.com/en-int/products/nexis-data/data-sets>.

⁸The word choice is deliberately broad because the phrase “islamic terrorist” appears to capture what is searched most often during a terrorist attack carried out or attributed to an Islamic group.

Figure 2 illustrates the time series for the raw and normalised number of articles and the Google search index. It is clear that all three are highly correlated, and that their highest spikes correspond to major terrorist attacks, also reported in the figure. Conceptually, the terrorist attacks are the real-world shocks that underlie any changes to either the number of articles published or Google searches made. The pair, in turn, causes changes to salience. Hence, in the months when terrorist attacks take place, the salience of Muslim communities peaks. It is then these changes in salience, that potentially cause violent backlash against local Muslim communities.

Figure 2: Plot of Muslim Salience



Notes: A plot of different measures of Muslim salience. The first is the number of UK-based newspaper articles written about the topic of Islamic terrorism (identified using several keyword combinations) as well as its associated index, the second is a relative index for the number of Google searches occurring in the UK about Islamic terrorism. The vertical dotted lines correspond to a selection of major Islamic terrorist attacks.

2.3 Police Reported Crime

The data on violent crime at the neighbourhood-level (34,750 LSOAs) is obtained from police crime records for England and Wales.⁹ The analysis focuses on violent crime, which is an aggregate measure based on the sum of violent & sexual offences, robbery, arson and criminal damage, possession of weapons, public order and disorder incidents, all of which are recorded at a monthly frequency for the 2011-19 period. Changes in Home Office Counting Rules (HOCR), that took effect in April 2018, for violence against persons and crimes relating to stalking and harassment, imply that after that date violent crimes are no longer comparable to the rest of the data. Moreover, prior to September 2011, the ‘arson and criminal damage’ incidents were part of the ‘other crime’ category, and were not recorded independently. Hence, I restrict the estimation sample to violent crimes observed from September 2011 to April 2018.¹⁰

Based on Table 2 below, on average, just under a third of all reported crime incidents are violent crimes, with about 2.5 per 1000 residents being reported per month in each

⁹The raw historical data is available at <https://data.police.uk>. It is compiled by 43 Police Force Areas (PFAs) that are responsible for the day-to-day running of policing activities. Data from the British Transport Police has been excluded due to potential measurement error associated with the location of the reported crimes. Data is not available for all LSOAs that fall into the Manchester PFA after July 2019 due to misreporting.

¹⁰Note that it is difficult to modify the analysis to extend beyond this date due to changes in the definition that imply that the new measure captures a fundamentally different dimension of violent crime. When looking at the Israeli-Palestinian conflict, I do use data from 2022-2024 where only the new definitions are in use.

neighbourhood. The distribution of violent crime is right-skewed; violence is ubiquitous in several neighbourhoods. The rate of violent crime is nearly 2 times higher in treated (4.50) compared to untreated (2.42) neighbourhoods. For identification, accounting for unobserved differences across neighbourhoods will be important to avoid endogeneity problems.

2.4 Additional Data

For the heterogeneity analysis and re-weighting exercises, I use a set of control variables defined at the neighbourhood and constituency-levels from the 2011 Census, available at the NOMIS repository. These characteristics are: population density, level of education, share of residents living in social housing, unemployment, youth share, religious fractionalization and share of foreign-born residents. I also include a per person estimate of losses from local austerity cuts, taken from Beatty and Fothergill (2013), as areas subject to deeper austerity cuts may subsequently experience a greater increase in crime in response to shocks to salience (Giulietti and McConnell, 2020).

In addition, to study the role of political preferences in shaping the degree of backlash, I collect data on election results at the constituency-level from the UK Electoral Commission. In particular, I use the results from the 2010, and 2015 General Elections and the 2016 United Kingdom European Union membership Referendum focusing on the share of votes in each constituency made in favour of right-wing or anti-immigrant parties. For the 2010 election, I focus on the combined vote for UKIP (UK Independence Party) and the BNP (British National Party) that, at the time, were the most popular parties advocating for strong restrictions to immigration. For the 2015 election, due to the fall of BNP into relative obscurity, I just focus on UKIP that absorbed most of the former’s supporters by that time. For the 2016 Brexit referendum, I use the share of votes supporting *Leave*.¹¹

Table 2 presents the mean and standard deviation of a selection of these control variables split by “treated” and “untreated” neighbourhoods. In general, “treated” units were located in constituencies that were more densely populated, had higher levels of education, had higher rates of religious fractionalization and a greater number of migrant residents, but were more strongly affected by austerity policies.

To discount alternative explanations such as improved reporting in violent crime over time, I collect NHS Hospital data on attendance and admission at Accident and Emergency (A&E) departments for serious injuries as a proxy for the occurrence of violent crime.¹² There are approximately 180 such departments, the data for which is available at a monthly frequency from July 2015. I consider the A&E departments within 500m of the nearest treated neighbourhood to be exposed.

¹¹In general, all three variables can be seen as proxies for the extent to which locals hold anti-migrant beliefs. As a result, they are strongly correlated ($\rho \geq 0.7$). Notably this implies, *contra* Fetzer (2019) that such views have roots before the start of any austerity policies.

¹²The data is publicly available and can be accessed at: <https://www.england.nhs.uk/statistics/statistical-work-areas/ae-waiting-times-and-activity/>

Table 2: Summary Statistics - Control Variables

Variable	Treated	Untreated
Violent Crime	4.497 (4.901)	2.421 (3.313)
2010 GE	5.232 (3.373)	5.559 (2.914)
2015 GE	12.294 (5.958)	14.264 (5.621)
Brexit	49.633 (12.280)	53.327 (10.943)
Pop. Dens.	90.167 (50.392)	35.406 (50.964)
Educ.	0.126 (0.025)	0.155 (0.030)
Relig. Frac.	0.576 (0.126)	0.465 (0.101)
Foreign	0.350 (0.141)	0.119 (0.127)
Austerity (GBPpc)	541.384 (91.180)	471.085 (120.646)

Notes: This table of summary statistics presents the mean and standard deviation (in brackets) for a selection of control variables split by treatment and control units. The variables are: share voting for UKIP/BNP in 2010 general election, share voting for UKIP in 2015 general election, share voting for Brexit, population density, share of residents with at most GCSE-level qualifications, religious fractionalisation, share of foreign-born residents, and austerity losses in GBP per person. The election results are at the constituency-level, the austerity losses are at the district-level, while the rest are at LSOA-level.

As an external validity exercise, I assess whether the salience-crime link holds for other ethnic or religious groups. In this case, whether the escalation of the Israeli-Palestinian conflict after October 2023 has led to violence being perpetrated against Jewish communities in England and Wales. Compared to Muslims, the number of Jews residing in the UK is very small standing at around 0.5% of the population in 2021, and largely concentrated in London. This means that the previous approach to measuring exposure will be less effective - there are simply too few Jewish residents elsewhere in the country. For this reason, I consider a neighbourhood as ‘treated’ if it is in the 99th percentile for the share of Jewish residents countrywide.¹³ The salience of Jewish communities is defined in an analogous way to above, using the intensification of the conflict, Google searches and Newspaper articles as the three key measures.¹⁴

¹³In this case, I do not consider the presence of a synagogue simply because the number is too small (fewer than 500 broadly-defined places of worship) to form an appropriately-sized treatment group. Hence the condition based only on the share of residents is used instead.

¹⁴For the newspaper articles, the following combination of terms is used: israel OR palestine OR israel-palestine OR gaza OR israeli-palestinian conflict OR gaza war OR hamas. These are the terms frequently used in conjunction with the intensification of the conflict in October 2023.

3 Empirical Methodology

3.1 Baseline

My primary goal is to identify the impact of changes in Muslim salience on the local violent crime rate at the neighbourhood level.¹⁵ If a rise in Muslim salience causes some degree of violent backlash, we should observe a larger increase in violent crime in areas with a visible and relative large Muslim community following a Muslim-led terrorist attack. As documented in Table 2, there is considerable heterogeneity in the characteristics of treated and untreated neighbourhoods. I thus implement a matched difference-in-difference design (Abadie, 2005). In the first step, a sample of treated and untreated units is constructed by matching on the basis of a relevant set of observables (discussed in more detail below); in the second step, a difference-in-difference model is run on this matched sample. The assumption is that the parallel trends assumption, necessary for the difference-in-difference analysis, holds conditional on the set of observables used in the matching step. Since there are no differences in the timing of shocks to Muslim salience, the difference-in-difference analysis can be conducted using a simple two-way fixed effects model given in equation (1):

$$y_{it} = \tau(\text{Salience}_t \times \text{Muslim}_i) + \alpha_i + \eta_t + \varepsilon_{it} \quad (1)$$

where y_{it} is the crime rate in neighbourhood i at time t , Salience_t is a variable measuring the salience of Muslim communities nationally at time t , Muslim_i measures whether a given neighbourhood is within 500 metres of a mosque and has a population of Muslim residents in the top 75th percentile of its local authority district, α_i and η_t are the neighbourhood and time fixed effects respectively, and ε_{it} is the error term.¹⁶ As discussed above, there are three measures of salience: the occurrence of major terrorist attacks, the normalised relative number of articles referencing Islamic terrorism, and a relative measure of the number of Google searches on the same subject. Importantly, as will be shown below, the terrorist attack index is extended not only to cover the month of the attack itself but also the subsequent five months to be consistent with the event study results. Note that both the newspaper and Google search measures have been discretised as dummy variables for whether the quantity of publications or searches is above the 75th percentile of the series. Different specifications of these variables will allow me to pin down both the extensive and intensive margin of the salience-crime relationship. Note that due to the initial matching step, standard errors need to be clustered at the level of the matched sub-group (Abadie and Spiess, 2022).

The average treatment effect on the treated (τ) is identified as long as counterfactual trends in crime are parallel between areas with and without visible Muslim communities:

$$E[\text{Muslim}_i \times \varepsilon_{it} | \alpha_i, \eta_t, \text{Salience}_t] = 0 \quad (2)$$

In the full sample, this assumption is not plausible as treated neighbourhoods differ substantially in terms of observable outcomes from untreated ones, which may also imply differences in crime trends. Hence, before estimating equation (1), it is necessary to select an appropriate set of control units.

As noted above, my main approach uses propensity score matching with a 1 : 3 ratio of

¹⁵See Appendix 5 which contains Figures B.1 and B.2 alongside TableB.1 that present the results for the baseline analysis for anti-social behaviour (ASB) but finds nothing. Violent crime is considerably more costly and is hence the focus of this paper.

¹⁶Note that I exclude from the estimation all neighbourhoods in which a terrorist attack happened, if it occurred in the UK.

treated and untreated units to construct such a sample. To do so, I use the standard nearest neighbour propensity score matching using a probit link function for seven key variables: average population growth, level of education (share of residents with at most a GCSE-level qualifications), proportion of youths (residents aged 0-24), manufacturing share of employment, share of residents living in socially-rented accommodation (social housing), share of residents that are long-term unemployed, and degree of religious fractionalisation. These covariates cover the socio-economic structure of each neighbourhood without undermining the assumption of common support. Since crime is often driven by socio-economic conditions (instead of say religious identities), matching on the basis of these observables should be sufficient.

Figure A.1 in the Appendix illustrates the results of this procedure. A visual inspection reveals that although complete balance has not been achieved across all variables, it has been greatly improved relative to the full sample, and only for the share of young people is there a statistically significant gap between treated and control neighbourhoods. Moreover, there is a high degree of overlap for most of the distribution of propensity scores and a lack of selection of the treated neighbourhoods in the matched sample.

Although the conditional parallel trends assumption is fundamentally untestable, evidence in support of its plausibility can be provided by modifying the baseline model in equation 1 to account for both pre-trend and dynamic effects that estimate differences in crime trends between treated and untreated units before and after a shock to salience. Formally, I estimate equation 3:

$$y_{it} = \sum_{j=-6}^{-1} \tau_j 1\{t = \gamma + j\} + \sum_{j=0}^{12} \tau_j 1\{t = \gamma + j\} + \alpha_i + \eta_t + \varepsilon_{it} \quad (3)$$

The main difference from 1 is the presence of 6 pre-treatment and 12 dynamic effect estimates which are stated relative to shock to salience occurring at time γ . All periods more than 6 months before each shock are used as the reference group. Importantly, the set of dynamic effects takes priority over their pre-treatment counterparts to avoid contamination. This means that a period that is within 12 months following an attack but less than six months before the next one is only used to estimate a dynamic effect, and not a pre-treatment coefficient. If the matching is successful, and the parallel trends assumption holds, then the coefficients for $j = [-6, -1]$ should be zero. newline

3.2 Heterogeneity Analysis and Extensions

This basic analysis is extended in a number of dimensions. Firstly, I document treatment effect heterogeneity across both neighbourhood and constituency-level characteristics. Intuitively, socio-economic conditions can shape the emergence of prejudice and dispose locals to out-group violence. Given the UK context, particular attention is paid to austerity cuts and voting behaviour, a proxy for political beliefs. Secondly, as locals may feel more threatened if attacks are targeted at and hurt members of their community, I compare the magnitude of retaliatory responses based on whether the terrorist act occurred in the UK or overseas. Thirdly, I examine whether the effect spills over to other minority groups residing within 1.5km of the treated community, thereby assessing the extent to which heightened inter-group hostility is diffuse rather than narrowly targeted. Fourthly, to explore whether salience-induced hostility manifests in physical violence beyond that captured by police-recorded crime, I estimate the effect of terrorist attacks on Accident and Emergency admissions, specifically examining whether the incidence of major injury presentations rises in the aftermath of an attack. Fifthly, I as-

sess the external validity of the salience-crime mechanism beyond the context of Islamic terrorism by exploiting the intensification of the Israel-Palestine conflict as a source of quasi-experimental variation, examining whether rates of violent crime in the vicinity of Jewish communities rose in its immediate aftermath.

Finally, I use an alternative approach that relies on a different set of identification assumptions to corroborate my basic findings. Specifically, I use a design that relies mainly on within-neighbourhood comparisons to identify the effect of changes to salience on violent crime. Formally, I estimate the panel regression in equation 4:

$$y_{it} = \alpha_i + m_t + \tau(\text{Salience}_t \times \text{Muslim}_i) + \beta \text{Salience}_t + \text{trend}_{it} + \varepsilon_{it} \quad (4)$$

where I use month fixed effects and fit neighbourhood-specific quadratic crime trends (trend_{it}) instead of time fixed effects. This allows the effect of salience (β) to be identified. Standard errors are clustered at the neighbourhood-level.

The relevant treatment effect is τ which captures how neighbourhood-level crime responds to changes in salience between areas with a visible Muslim community and those without one. The specification in equation 4 bears a resemblance to both a difference-in-difference style design since comparisons are made between treated and untreated areas, and a Bartik-style shift-share design where Muslim_i (capturing cross-sectional variation in Muslim communities across England and Wales) measures local exposure to national shocks in salience, Salience_t . For the main specification, the parallel trends assumption is not required as I allow for differential trends in crime via the inclusion of neighbourhood-specific quadratic crime trends. More formally, I require that the composite term ($\text{Salience}_t \times \text{Muslim}_i$) is uncorrelated with any other determinants of the crime rate contained in the error term (ε_{it}). This implies that:

$$E[\text{Muslim}_i \times \varepsilon_{it} | \alpha_i, m_t, \text{Salience}_t, \text{trend}_{it}] = 0 \quad (5)$$

Hence, the biggest threat to identification in this case comes from variables codetermined with the distribution of Muslim communities that are correlated with changes in the crime rate (Adão et al., 2019; Goldsmith-Pinkham et al., 2020). Put another way, these are variables that would cause crime to rise when Muslim salience increases even in the absence of any Muslim residents. If the neighbourhood-specific quadratic time trends are adequately specified, then this assumption is not implausible, yet remains untestable. To bolster its credibility, I conduct a number of robustness checks and falsification exercises aimed at showing that shocks to salience affect only Muslim communities.¹⁷

Next, I consider whether neighbourhoods close to treated areas also see an increase in violent crime. I rely on within-neighbourhood variation and estimate geographical spillovers using equation 6, where neighbourhoods are divided into 500 metre bins based on their distance from a treated area. I use 10 such bins, so the maximum distance considered here is 5km:

¹⁷These are analogous to the ones performed for the matched difference-in-difference approach. To start with I consider whether areas with large populations of other religious minorities, namely Jews, Hindus and Sikhs, experience a similar growth in crime during times of higher Muslim salience. In addition, I make sure that the results are not driven by the definition of a treated area.

$$y_{it} = \alpha_i + m_t + \tau(\text{Salience}_t \times \text{Muslim}_i) + \sum_{j=1}^{10} \gamma_j(\text{Dist}_j \times \text{Salience}_t) + \beta \text{Salience}_t + \text{trend}_{it} + \varepsilon_{it} \quad (6)$$

Most variables have the same interpretation as before. However, the variable Dist_j is a binary indicator for whether a given neighbourhood i belongs to bin j , and the parameter γ_j measures the effect of changes to Muslim salience for each distance.

Finally, I use this corroborative approach to revisit and explore more deeply the question of whether voting behaviour shapes the response of violent crime to shocks to salience. The larger sample size allows me to check whether variation in the magnitude of responses can be attributed to compositional differences in characteristics across neighbourhoods.

3.3 Robustness Checks

In support of the baseline results, I conduct a battery of robustness checks. Firstly, I consider two alternative control unit samples. The first is the set of otherwise untreated neighbourhoods that are within 500m of a treated area. The intuition behind this choice is that since both neighbourhoods are very close, they likely face similar kinds of shocks, and both have sufficiently similar characteristics. This makes it plausible to think that these units would behave similarly in counterfactual terms as well. The main limitation of these units, is the possible presence of geographical spillover effects, that would potentially bias results. The second set of control units is defined as neighbourhoods with a large Muslim minority and a place of worship that is not a mosque (e.g. a prayer or multi-faith room). These neighbourhoods also contain a relatively large number of Muslim residents, but as there is no visible Mosque present, it would be more difficult to target these communities during times of greater salience. However, if the factors that explain the absence of a mosque in these neighbourhoods are correlated with unobservable determinants of crime, the parallel trends assumption would be violated.

Secondly, I re-estimate my baseline model for 1:1 and 1:5 matching ratios. Moreover, I check whether altering the set of covariates affects the results. Specifically, I add population density, share of residents born outside the UK, share of residents working in manual or semi-skilled occupations, gross value added, share of residents that are male, and share of residents in full-time work as additional variables in the matching procedure. These were not selected in my preferred specification because of issues with the common support assumption: the overlap in propensity scores between treated and untreated units deteriorates considerably in the full sample. This occurs because Muslim communities tend to have a high number of family members born abroad, that may not be in full-time work or are potentially employed (even if temporarily) in low-skilled jobs. Hence, the addition of these economic variables risks creating a separation between treated and untreated areas.

Thirdly, I consider whether the results are simply driven by the definition of a treated area. Hence, I modify the treatment definition to either: (i) whether the share of Muslim residents is in the 90th percentile for the local authority district, or (ii) whether the share of Muslim residents is an outlier, defined as a share greater than $1.75 \times \text{IQR}$ (where IQR is the interquartile range), for the local authority district.

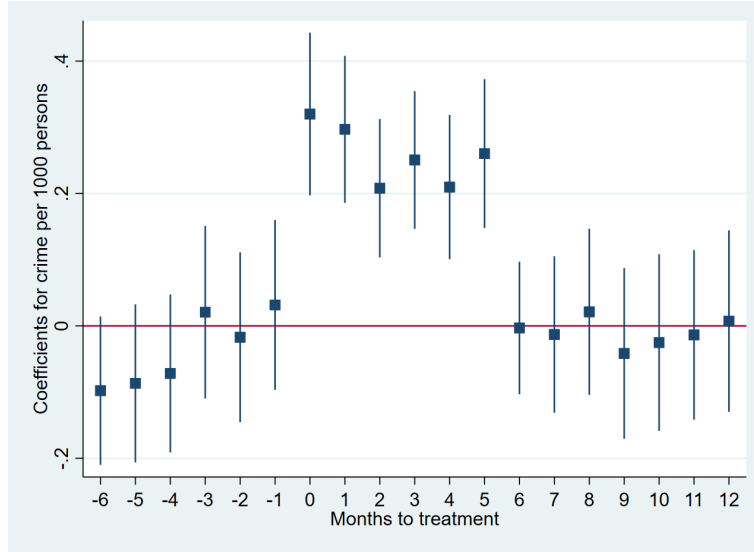
Fourthly, as an important falsification exercise, I consider whether areas with large populations of other religious minorities, namely Jews, Hindus and Sikhs, experience a similar growth in crime during times of higher Muslim salience. I expect that these effects should not be statistically significant. Lastly, I tentatively check for whether the results are driven by the presence of spatial spillover effects that may create bias in difference-in-difference estimates through violations of the SUTVA condition.

4 Results

4.1 Muslim Salience and Crime

The baseline event study estimates of the impact of an Islamic terrorist attack are presented in Figure 3.¹⁸ These are estimated on the 1:3 propensity score matched sample, the preferred choice.¹⁹ The estimates show a strong increase in violent crime of nearly 13% relative to the average.²⁰ This persists for 6 months before returning to its pre-attack level. Reassuringly, both in the pre-period and in the months after the dynamic effects dissipate, violent crime appears to follow the same trends in both the treated and control units. This supports the plausibility of the parallel trends assumption.

Figure 3: Event Study Plot for Violent Crime



Notes: An event study plot for the effect of Islamic terrorist attacks on the rate of violent crime. Estimation is based on a 1:3 propensity score matched sample. A 95% confidence interval is included around each estimate.

¹⁸For the full set of event study plots covering all three salience measures, see Figure A.2 in the Appendix.

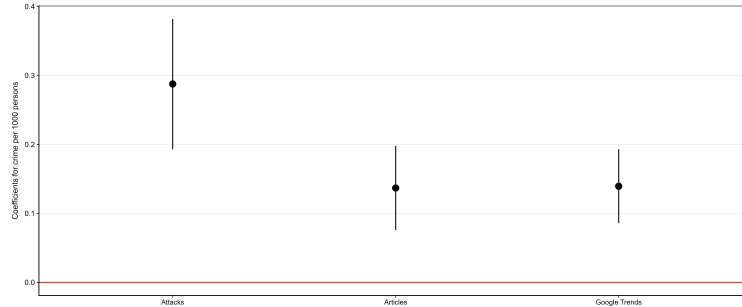
¹⁹See the section 4.4 for a discussion of the other possible samples.

²⁰In interpreting these coefficient estimates, I will commonly refer to percentage changes in the crime rate. Unless stated otherwise, these will be relative to the *mean* of the monthly violent crime rate at the neighbourhood-level over the entire sample period, corresponding to a measure of a long-run national level of violent crime.

As noted above, three different measures of Muslim salience are used: major terrorist attacks, the relative number of newspaper articles written about Islamic terrorism and a relative measure of the number of Google Searches that link Islam and terrorism. The first reflects the actual events that make Muslim communities more salient to the general public, while the last two are more noisy reflecting the supply of and demand for information, respectively. Apart from illustrating two potential channels through which salience leads to an effect, the inclusion of these additional measures acts as a robustness check for whether any changes in crime are really driven by acts of terrorism or other national-level shocks.

Figure 4 presents the average treatment effect estimates using all three of these measures. To calculate it, both the newspaper and Google search measures have been discretised as dummy variables for whether the normalised quantity of publications or searches is above the 75th percentile of the series. The terrorist attack index has been extended not only to cover the month of the attack itself but also the subsequent five months to be consistent with the event study results. It shows that over the six months following an attack, on average, violent crime rises by about 11.5%. This again confirms that the retaliatory response to changes in salience is both large and significant. The impact as measured through newspaper publications and Google searches is smaller in magnitude with both estimates showing a 5-6% increase.²¹ This suggests that news coverage, and hence the supply of negative information, is as important for violent behaviour as demand for information about the attacks.

Figure 4: Treatment Effect Plot for Violent Crime



Notes: A plot of estimates for the average treatment effect on the treated (ATT) of three measures of Muslim salience on crime. Estimation is based on a 1:3 propensity score matched sample. A 95% confidence interval is included around each estimate.

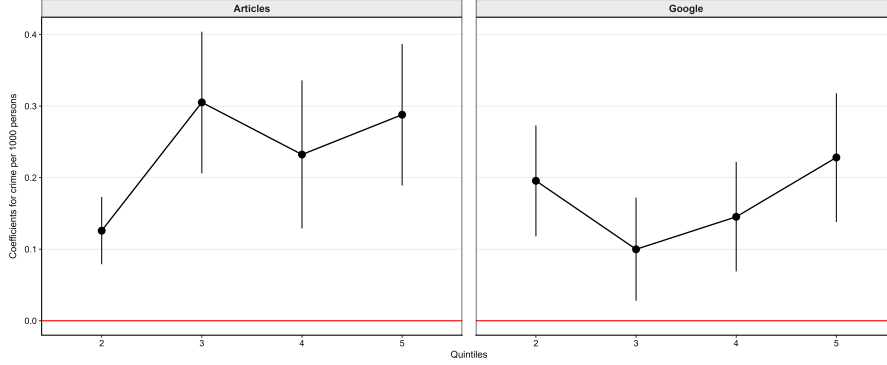
In addition to the ATT estimates that represent the extensive margin of a shock to salience, the measures based on newspaper articles and Google searches can be used to study how the effect on crime varies with salience intensity. To do so, I compare the ATTs along different quintiles of article publications and searches using the first quintile as the reference value. Figure 5 presents the estimates for each quintile.²² For published newspaper articles, there is a clear monotonic relationship between salience intensity and crime: at the second quintile the effect is about 4%, but rises to nearly 12% at the fifth

²¹Note that the difference in effect sizes may be explained by the choice of cut-off for the discretisation of the two measures, and likely does not reflect an upward bias in the attack estimates.

²²Following Callaway et al. (2024), I do not consider the marginal treatment effect of moving from one quantile to the other, instead simply estimating level effects that compare a given quantile of treatment to its absence, or in this case, a reference level. Since I am not using a subset of either control or treatment units, the necessary identification assumptions are largely unchanged from the baseline approach.

quintile. For Google searches, the association is less clear with the treatment effect rising to at most 9%. This result underscores the importance of the media for propagating information about attacks, and potentially exacerbating retaliatory responses through reporting choices.²³

Figure 5: Intensive Margin of Muslim Salience Plot



Notes: A plot of estimates for the average treatment effect on the treated (ATT) for newspaper articles and Google searches along the intensive margin. For both measures their continuous values are split by quintiles and the first is used as the reference group. Estimation is based on a 1:3 propensity score matched sample. A 95% confidence interval is included around each estimate.

4.2 Heterogeneity Analysis

4.2.1 Basic Characteristics

A natural question for policymakers is: which local area characteristics strengthen or dampen the salience-crime link? I attempt to answer this question at two different levels: neighbourhood and constituency. While the former is directly relevant for capturing neighbourhood differences, the second helps capture the spatial interrelationships between neighbourhoods that shape treatment effect heterogeneity. For this reason, I conduct heterogeneity analysis based on a wide range of social and economic characteristics comparing treatment effects for neighbourhoods above or below the median for each one.²⁴ Table 3 presents the results for the neighbourhood-level (LSOA-level) characteristics.

²³It is unlikely that this association is explained by reverse causality, i.e. local violence leading to a greater number of articles or searches, because the publicity is more strongly correlated with the scale and intensity of the original terrorist attack.

²⁴For an explanation of each variable see Table A.1 in the Appendix.

Table 3: Heterogeneity Analysis - LSOA-level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Treat	0.216 (0.239)	0.341*** (0.052)	0.345*** (0.068)	0.337*** (0.115)	0.790*** (0.257)	-0.124 (0.097)	0.305 (0.401)
Treat x Pop Dens.	0.079 (0.244)						
Treat x Educ.		-0.121 (0.144)					
Treat x Soc. Rent			-0.079 (0.090)				
Treat x Unemp.				-0.109 (0.125)			
Treat x Prop. Youth					-0.541** (0.262)		
Treat x Relig. Frac						0.462*** (0.112)	
Treat x Foreign							0.061 (0.404)
P	0.000	0.000	0.000	0.000	0.000	0.000	0.000
N	433326	433326	433326	433326	433326	433326	433326

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the LSOA-level heterogeneity analysis estimates for the average treatment effect on the treated of changes to Muslim salience on the monthly crime rate at the LSOA-level. Each model uses time, time by characteristics and neighbourhood fixed effects in a triple difference-in-difference-style design. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the matched sub-group level.

There is little heterogeneity in treatment effects by social characteristics. The two exceptions are the proportion of youth and religious fractionalization: having more than 30% of local residents being between the ages of 0-24 reduces the treatment effect from about 32% to only 10%. This can be partly attributed to the fact that victims of violent hate crimes are often older individuals above 30 (Walters and Krasodomski-Jones, 2018). In addition, neighbourhoods with higher religious fractionalization see violent crime rise by just over 13%, with the effect disappearing completely in low fractionalization neighbourhoods. Indeed, fractionalization has been linked to greater tensions among residents that can lead to more violent outcomes, especially following a terrorist attack (Alesina and La Ferrara, 2005). In addition to analysing the role of neighbourhood-level characteristics, I perform a similar analysis using constituency-level covariates, as the wider socio-economic environment around a neighbourhood may shape criminal behaviour. Table A.2 presents the results, showing no significant significant effects across any variable. The degree of backlash against Muslim communities varies little with the characteristics of either the neighbourhood or wider constituency.

4.2.2 Political Preferences

Personal attitudes and prejudices may shape both violent behaviour and voting, which suggests the treatment effect may vary across political preferences, the revealed preferences of local residents through the voting outcomes at general elections or referenda (Colussi et al., 2021). Specifically, I check whether violent crimes respond more strongly in neighbouring areas nested within constituencies whose share of support for right-wing parties (RW, comprising UKIP and BNP) or leaving the EU was above the median. The results of this test are collected in Table 4. I also report results for the measures of salience based on newspaper articles and Google searches as these are important media through which political beliefs are formed.

Table 4: Heterogeneity Analysis - Voting Behaviour

	(1) Attack	(2) Attack	(3) Attack	(4) Art.	(5) Art.	(6) Art.	(7) Google	(8) Google	(9) Google
Treat	0.230*** (0.054)	0.202*** (0.048)	0.243*** (0.052)	0.098*** (0.036)	0.100*** (0.033)	0.130*** (0.035)	0.110*** (0.032)	0.107*** (0.029)	0.124*** (0.031)
T x RW	0.093 (0.099)			0.068 (0.064)			0.051 (0.057)		
T x UKIP		0.259** (0.106)			0.115* (0.069)			0.100* (0.061)	
T x Brexit			0.079 (0.104)			-0.003 (0.067)			0.023 (0.059)
p	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
N	433326	433326	433326	433326	433326	433326	433326	433326	433326

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the LSOA-level heterogeneity analysis estimates, in terms of political preferences, for the average treatment effect on the treated of changes to Muslim salience on the monthly crime rate at the LSOA-level. Each model uses time, time by voting share and neighbourhood fixed effects. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. The models test the difference in effects between being below or above the median in terms of voting shares for either right-wing parties or Brexit. Standard errors are reported in parentheses and are clustered at the matched sub-group.

In all cases, areas with higher support for anti-immigrant parties saw, on average, a larger violent backlash. However, the effects are only significant for the 2015 General Election. Areas that supported UKIP saw violent crime jump by nearly 10% more in response to a terrorist attack than those that did not. Similarly, the interaction term is weakly significant for published articles and Google searches showing a roughly 5% larger rise in violent crime. This complements the evidence in Clifton-Sprigg et al. (2020), who find a 15-20% increase in hate crimes after the Brexit referendum. More generally, these results highlight how the progressive normalisation of private prejudices can increase willingness to express these views violently against minorities (Bursztyn et al., 2020; Rushin and Edwards, 2018). I revisit this issue below, and try to check whether this gap can be explained by compositional differences in the characteristics of the different constituencies.

4.2.3 Austerity

Following the 2008 Financial Crisis, the UK implemented large welfare cuts to reduce the government budget deficit. A recent and important literature has found a robust link between austerity policies and local crime (Bray et al., 2024; Facchetti, 2025; Giuli-etti and McConnell, 2020; Villa, 2024). Moreover, it is possible that those most strongly affected by the austerity policies could attribute their economic woes to migrants and other minorities, leading to stronger backlash in neighbourhoods most affected by aus-terity.²⁵ Table 5 presents a triple difference-in-difference model, using a binary measure of above and below median per capita welfare losses from austerity policy (GBP) at the LAD-level.

²⁵A similar idea is proposed in Fetzer (2019) as an explanation for the results of the 2016 Brexit vote that resulted in the UK leaving the EU. This mechanism is suggested in Guriev and Papaioannou (2022) as an important explanation for the rise of populism across many European economies.

Table 5: Heterogeneity Analysis - Austerity

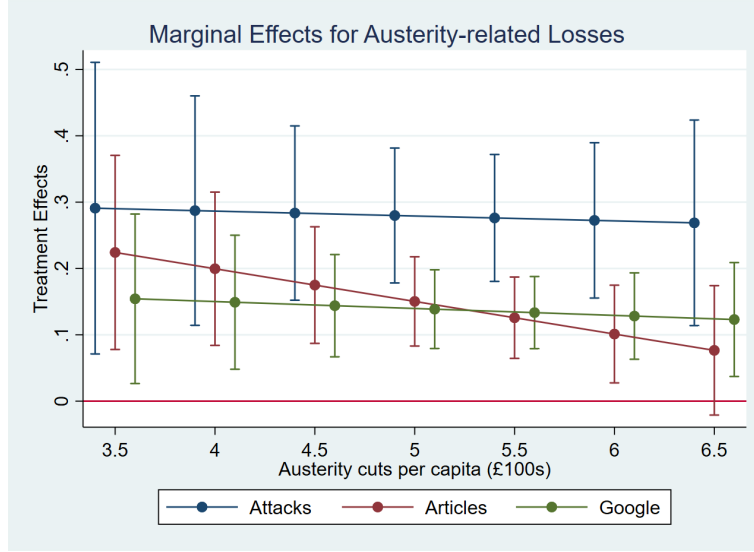
	(1)	(2)	(3)
Treat Attacks	0.345*** (0.111)		
Treat Attacks x Aust.	-0.092 (0.123)		
Treat Articles		0.212*** (0.075)	
Treat Articles x Aust.		-0.106 (0.082)	
Treat Google			0.180*** (0.066)
Treat Google x Aust.			-0.058 (0.073)
p	0.000	0.000	0.000
N	429861	429861	429861

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the LSOA-level heterogeneity analysis estimates for the average treatment effect on the treated of changes to Muslim salience on the monthly crime rate at the LSOA-level. Each model uses time, time by austerity loss and neighbourhood fixed effects in a triple difference-in-difference-style design. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the matched sub-group level.

Across all three measures of minority salience, the interaction term estimates are not statistically significant, suggesting that the connection between salience and violent crime does not vary across areas hit differently by austerity. To account for the possibility that the split along the median is too broad, I re-estimate the model but treat the per person losses as a continuous variable instead. Figure 6 below illustrates the estimated marginal effects on crime for per person austerity-related losses (£100s). Again the flat shape of the marginal effects curves suggests that there is little heterogeneity in this dimension.

Figure 6: Austerity Marginal Effects



Notes: A plot of estimates for the marginal effect of austerity cuts on average treatment effect on the treated (ATT) of Muslim salience on violent crime. Estimates are derived for all three measures of salience: terrorist attacks, newspaper articles and Google searches. A 95% confidence interval is included around each estimate.

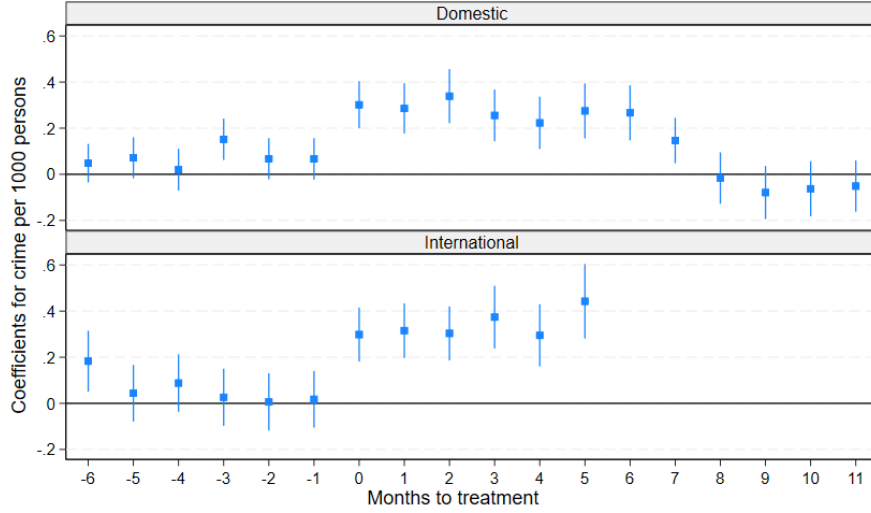
As a result, I do not find much support for the conflict theory, which predicts that austerity policies (or simply those that increase deprivation) can work to undermine social cohesion, by intensifying competition between different social groups (Anderson et al., 2020; Bray et al., 2024), hence challenging some previous empirical analysis linking worsening economic conditions and anti-minority crimes (Dustmann et al., 2011; Falk et al., 2011; Krueger and Pischke, 1997).

4.3 Extensions

4.3.1 Attack Location

An important question is whether there has been a larger response to domestic versus international terrorist attacks in terms of violent crime. The magnitude of estimates may give insights about the behavioural preferences of those willing to commit hate crimes, and whether threat perceptions vary by whether an attack happens close by or in another country (Alba et al., 2005). The event study plots for both types of attacks are illustrated in Figure 7 below.

Figure 7: Attack Location Effects



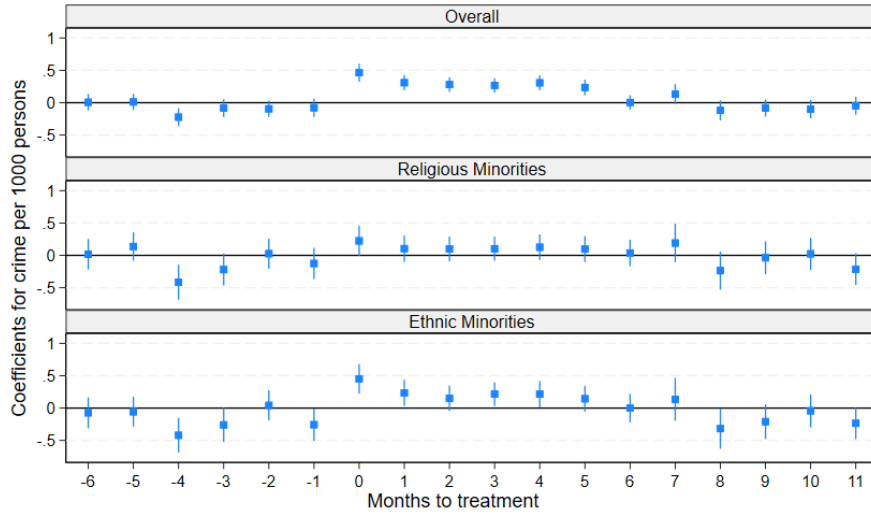
Notes: A set of event study plots of estimates of the dynamic effects of terrorist attacks, split by whether they occurred in the UK or elsewhere, on violent crime. A 95% confidence interval is included around each estimate.

The relative frequency of international attacks is higher meaning that I cannot identify as many clean post-treatment periods for such shocks. Nonetheless, the effects for both types of attack are relatively similar in magnitude corresponding to a roughly 12% increase in violent crime for the first six months. This may seem unexpected at first glance, however, many of the international attacks received considerable publicity and were typically responsible for a higher number of casualties than those occurring in the UK.

4.3.2 Minority Spillovers

Next, I consider another issue that has thus far been neglected in the salience literature, ethnic and minority spillover effects. The idea is that an increase in Muslim salience, and thus violence directed at the Muslim community, may affect violence towards other minority communities nearby. Many of those committing hate crimes may not distinguish between different minority groups, leading to spillover effects across ethnic and religious lines. To test this, I perform event study analysis on neighbourhoods in the top 90th percentile of their local authority in terms of the size of their ethnic (Mixed, Asian, Black, Other) and religious (Buddhist, Hindu, Sikh, Jewish, Other) minority populations that are within 1.5km of a ‘treated’ neighbourhood. The results are presented in Figure 8.

Figure 8: Ethnic and Religious Minority Spillover Effects



Notes: A set of event study plot of estimates for the dynamic effects of terrorist attacks on violent crime among ethnic and religious minority communities located within 1.5km of a ‘treated unit’. A 95% confidence interval is included around each estimate.

The overall results indicate a clear spillover effect that is largest in the month of the attack, showing a jump in violent crime by around 20%, lasting about 6 months, a pattern similar to the one observed for Muslim minorities. Most of the effect is driven by spillovers to ethnic minorities, not religious ones. With the exception of the month of the terrorist attack itself, violent crime remains stable in neighbourhoods with religious minorities. This implies that policy prescriptions should focus on correcting prejudices aimed at immigrants, as well as racial biases, that may be the source of these retaliatory responses. Violent crimes may not be motivated by feelings of wrongdoing but that, following a terrorist attack and after witnessing violence against Muslims, those with a higher propensity towards such crimes may feel more willing to express and act on their existing prejudices (Clifton-Sprigg et al., 2020). This complements the findings above that areas with higher degrees of fractionalization see significantly larger responses in crime.

4.3.3 Israeli-Palestinian War

Much of the salience literature has focused on Muslim communities in the West. This is primarily because these are large minority groups that have often been portrayed negatively in the media, and especially by many populist commentators and politicians. It is important to assess whether salience can affect the outcomes of other groups.

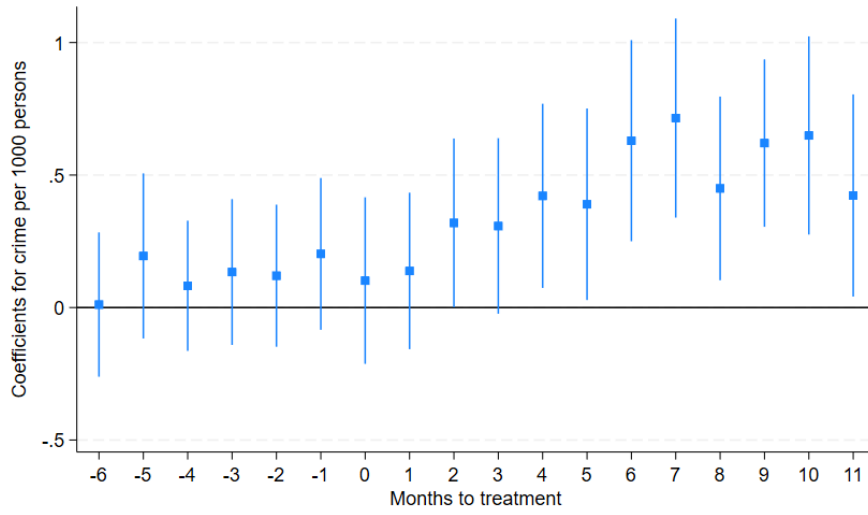
To test this, I consider the impact on Jewish communities following the intensification of the Israeli-Palestinian Conflict in October 2023 as a result of the Hamas attack on Israel. Subsequently, the UK experienced a surge in anti-Semitic hate crime where the number of incidents far exceeded historic averages.²⁶ To do so, I use a similar matched

²⁶For more information see the reports by the European Jewish Congress, <https://eurojewcong.org/news/communities-news/united-kingdom/cst-records-all-time-high-in-antisemitic-incidents-in-the-uk-after-october-7/>, and the Community Security Trust (CST), <https://cst.org.uk/research/cst-publications/antisemitic-incidents-january-june-2026>.

difference-in-difference design with 1:5 nearest neighbour matching.²⁷

Figure 9 presents the event study estimates for the conflict itself. Although there is no initial growth at the start of the conflict, violent crime appears to rise several months later. This is consistent with the timeline of the conflict, where the response by the Israeli military accelerated in the winter months. The effects are large and significant with a peak increase in violent crime of nearly 26% relative to the baseline.

Figure 9: Event Study Plot for Violent Crime in Jewish Communities



Notes: An event study plot for the effect of the intensification of the Israeli-Palestinian conflict on the rate of violent crime in Jewish communities. Estimation is based on a 1:5 propensity score matched difference-in-difference approach. A 95% confidence interval is included around each estimate.

Table 6 presents the full set of average treatment effects on Jewish communities along both the extensive and intensive margins. The static effects indicate that in Jewish communities the conflict has caused an increase in violent crime of about 13-16%, rising to nearly 17.5% when salience is at its highest. Overall, this supports the contention that salience is a powerful motivator of human behaviour, and explains the diffusion of violent crimes. Equally, it supports the idea that the salience-crime link holds generally across society, thus presenting a challenge for policymakers that wish to balance media freedom and social harmony.

²⁷The plots illustrating common support and showing covariate balance before and after matching are given in Figure A.4 in the Appendix. The slightly higher ratio is preferred here because there are a smaller number of Jewish communities in the UK, and hence a larger number of similar ‘control’ units available.

Table 6: Jewish Salience and Violent Crime

	Attack (1)	Articles (2)	Google (3)	Articles (4)	Google (5)
Treat	0.372*** (0.103)	0.386*** (0.091)	0.342*** (0.088)		
Q2				-0.139 (0.093)	0.035 (0.066)
Q3				0.013 (0.077)	-0.023 (0.077)
Q4				0.226*** (0.087)	0.271** (0.114)
Q5				0.391*** (0.095)	0.433*** (0.113)
p	0.000	0.000	0.000	0.001	0.002
N	66078	66078	66078	66078	66078

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

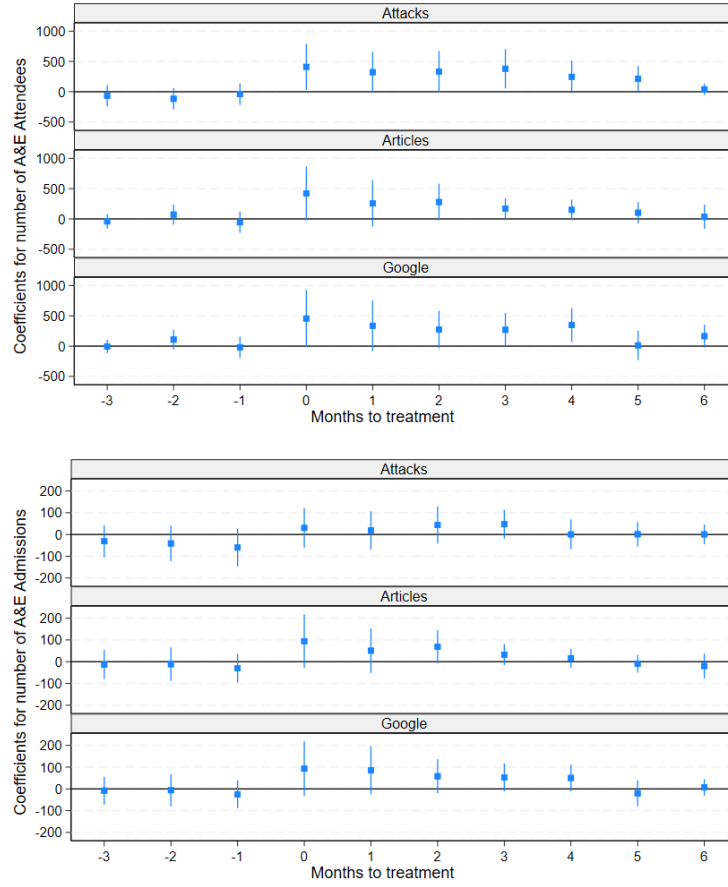
Notes: This table reports the baseline 1:5 matched difference-in-differences estimates for the average treatment effect on the treated of changes to Jewish salience on the monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effects. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. Three measures of salience are used: the start of the Israel-Palestine conflict, newspaper articles and the Google Trends search index. Models (1)-(3) consider effects at the extensive margin using simple binary measures of salience, while Models (4)-(5) instead estimate effects on the intensive margin using the number of articles and relative search intensity as salience measures. Standard errors are reported in parentheses and are clustered at the matched sub-group level.

4.3.4 Accident and Emergency

An important alternative explanation of my results is that they are caused by an increase in reporting of offences. That is, violent crime does not rise but when salience is higher Muslim populations become more likely to report the crimes they experience. Alternatively, police anticipating the possible retaliatory violence may expend more resources in the affected neighbourhoods, increasing the reporting of crime. In either case, the reported crime rate may increase, without any changes to actual crime on the streets. I therefore use accident and emergency (A&E) attendance and admissions, as a proxy measure for violent crime and its consequences. I compare the number of individuals either attending A&E departments or being admitted due to serious injuries, in hospitals close to (within 500m) treated units, relative to those further away.

Figure 10 illustrates the dynamic effect of a shock to salience on attendance and admissions. In the month of a terrorist attack, attendance of A&E increases significantly by about 411 persons per month corresponding to a nearly 8% increase over the average number. This effect size appears consistent across all three measures of salience, and with the effect size on violent crime. Importantly, the number of admissions, a proportion of the attendances where formal hospitalisation is required with additional inpatient care, is left largely unaffected, and the impact there is smaller and not statistically significant. Hence, the salience-crime link cannot be purely attributed to changes in detection or reporting behaviour: there is a genuine outbreak of violence around Muslim communities, but no increase in the severity of the violent crimes committed.

Figure 10: Event Study Plots for A&E Attendance and Admissions



Notes: An event study plot for the effect of an increase in Muslim salience on attendance and admissions to A&E departments due to serious injury in areas close to exposed neighbourhoods. A 95% confidence interval is included around the estimates.

4.3.5 Corroborative Approach

Instead of relying on between-unit comparisons, the corroborative approach primarily leverages within-neighbourhood variation to identify the average treatment effect on the treated, and, as such utilises the entire sample of neighbourhoods. Hence, if unbiased, this is more efficient than the matched difference-in-difference approach. Table 7 reports these estimates for each salience measure. Compared to the previous results, these estimates, though positive and mostly significant, are considerably smaller. They suggest that violent crime in ‘treated’ neighbourhoods rises by around 7% following a terrorist attack.²⁸

²⁸The two robustness checks of these results can be found in Tables A.3 and A.4. The alternative treatment definitions yield estimates consistent with their baseline counterparts. Curiously, the situation is more complex with the placebo minority estimates. I find that the shock to published articles is associated with a weak negative effect on Jewish communities, and a weak positive on Sikhs.

Table 7: Muslim Salience and Violent Crime Estimates

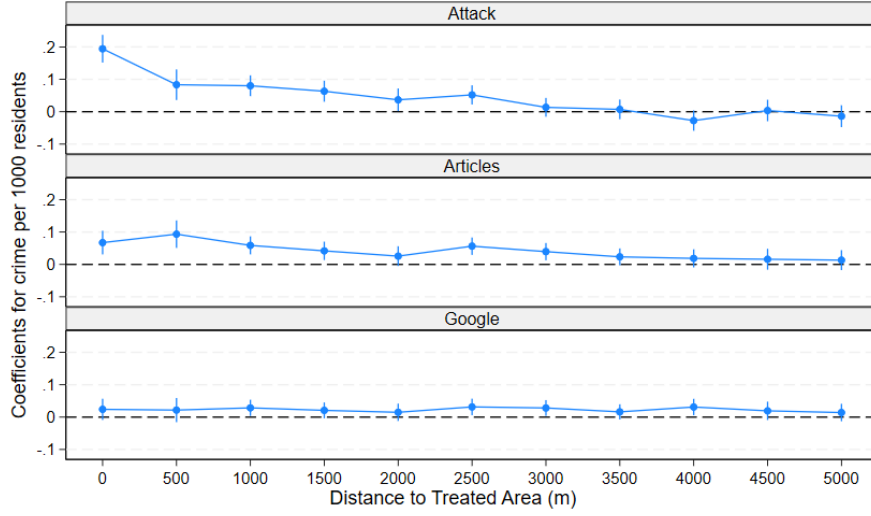
	(1)	(2)	(3)
Attack x Q75 Mosq.	0.177*** (0.022)		
Articles x Mosq.		0.048*** (0.018)	
Google x Mosq.			0.013 (0.017)
N	2675711	2675711	2675711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. Three measures of salience are used: terrorist attacks, newspaper articles and the Google Trends search index. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

The larger sample sizes make it easier to look at estimates across different sub-samples of the data. Hence, I revisit the issue of spatial spillovers and voting behaviour below. Figure 11 presents the estimates of equation 6 for the three main measures of salience. For the attacks themselves, there appears to be monotonic effect that diminishes with distance. That is, neighbourhoods closer to the treated areas also see a positive effect on violent crime. This becomes insignificant at the 3km mark. The relationship is similar for newspaper articles where the decline occurs substantially faster. If plausible, these estimates imply that the matched difference-in-difference results are, if anything, underestimates of the true treatment effect. Nonetheless, removing the control units within 1.5km of any ‘treated’ units had no discernible effect on the results. This may suggest that the units selected in the unconstrained matching step, though close in terms of distance, were simply those that were not strongly affected by these spillovers.

Figure 11: Neighbourhood Spillover Estimates



Notes: A plot of estimates for the average treatment effect on the treated (ATT) salience on violent crime. A 95% confidence interval is included around each estimate.

Next, I consider again voting behaviour. Above, I found evidence for the existence of a gap in how much violent crime jumps in response to a shock to salience between constituencies according to their degree of support for broadly right-wing or anti-immigrant political parties or agendas. The specific reasons for why this gap arises are not apparent. One important explanation for this heterogeneity is the existence of compositional differences in the characteristics of the different constituencies. For example, it could be that the degree of fractionalization influences both voting behaviour and strengthens the salience-crime, thus explaining the gap in effects. If the contributions of different types of variables could be identified, it would provide policy makers with the tools to promote greater social cohesion without sacrificing diversity.

To do so, I draw on the work of DiNardo et al. (1996) and more recently Illing et al. (2024) in calculating a “composition-adjusted gap” in treatment effects between neighbourhoods located in constituencies whose share of votes was above/below the median for right wing parties (2010), UKIP (2015) and Brexit (2016), all through a re-weighting strategy.²⁹ As suggested above, the difference in estimated effects observed above could be explained by differences in the distribution of observable outcomes between the two types of constituencies. To overcome this challenge, I re-weigh the constituencies that voted against right-wing parties/UKIP/Brexit (*low voting*) to match the distribution of observables in the constituencies that voted for them (*high voting*).

Formally, this approach can be summarised following Illing et al. (2024). Suppose that τ is the treatment effect for the impact of a terrorism-induced change to Muslim salience on violent crime for any given neighbourhood. Then the raw unadjusted gap, observed above, can be written as the difference in average treatment given the corresponding constituency’s voting behaviour:

²⁹Note that this approach does not affect the validity of the matched difference-in-difference design as the samples of treated and untreated observations remain intact. The approach here performs the re-weighting across the voting dimension of the units.

$$Gap_{unadj} = E[\tau|High] - E[\tau|Low] \quad (7)$$

Now consider a vector of covariates X which are potential determinants of the size of the gap and vary between constituencies. In order to account for differences in composition, we would like to find the treatment effect of neighbourhoods in low voting constituencies adjusted to the covariate distribution of high voting ones:

$$E[E[\tau|Low, X]|High] \quad (8)$$

To achieve this, again following Illing et al. (2024), I use a re-weighting function, $\phi_x(x)$, to map the characteristics of low to high voting constituencies. This then yields a composition-adjusted gap:

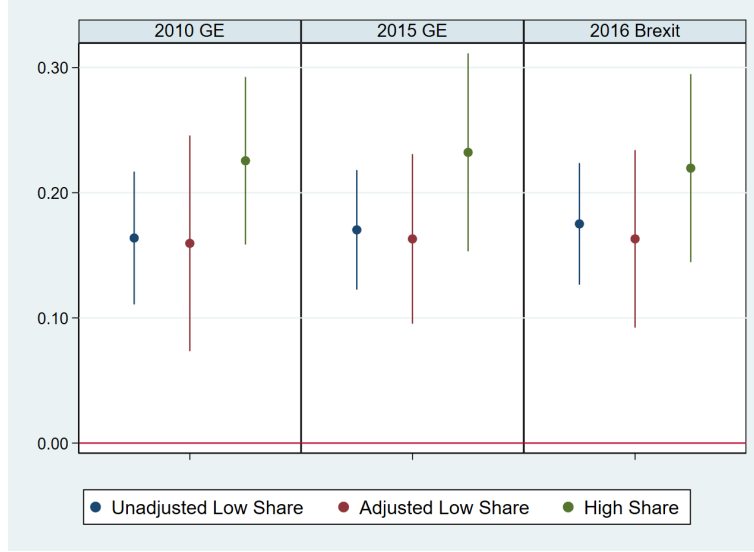
$$Gap_{adj} = E[\tau|High] - \int_{\mathcal{X}} E[\tau|Low, X] dF_X^{Low}(x) \phi_x(x) \quad (9)$$

In practice, I use a probit model to estimate the propensity score, given a set of social and economic observables, of each constituency having a high or low vote share for right-wing parties, UKIP or Brexit. This is done at the constituency-level, since this is the level at which voting shares are reported. A major concern is the failure of the overlap condition. For example, in the UK, individuals and areas with lower educational attainment were more likely to support anti-immigrant parties and vote for Brexit (Alabrese et al., 2019; Becker et al., 2017, 2018). This however introduces a near separation of the constituencies based on their vote shares. Hence, to avoid common support problems, I use principal components analysis to reduce the number of dimensions of the data. The characteristics used are: population density, proportion of residents aged 0-24, religious fractionalisation, proportion of foreign-born residents, change in proportion of foreign-born residents over the preceding decade, share of residents born in Africa or the Middle East, manufacturing share of employment, share of residents living in socially-rented accommodation (social housing) and share of residents in full-time work. These are reduced to two principal components that explain over 75% of variance of the covariate set.³⁰ Neighbourhoods in high-voting constituencies are then assigned a unit weight, while those in low voting constituencies receive a weight of $\frac{\hat{p}}{1-\hat{p}}$, where $\hat{p}$ is the estimated propensity score.

The results of this re-weighting exercise are presented in Figure 12. Although the estimates are not very precise, an important conclusion can still be gleaned from the results: most of the gap in backlash across political preferences cannot be explained by social, economic and demographic factors. For example in the 2015 General Election result, where the gap was previously largest, the coefficient on the treatment variable, corresponding to the effect on violent crime in areas with a low share, does not change whether re-weighting or not.

³⁰See Figure A.3 for the relevant scree plot.

Figure 12: Voting Shares and Violent Crime



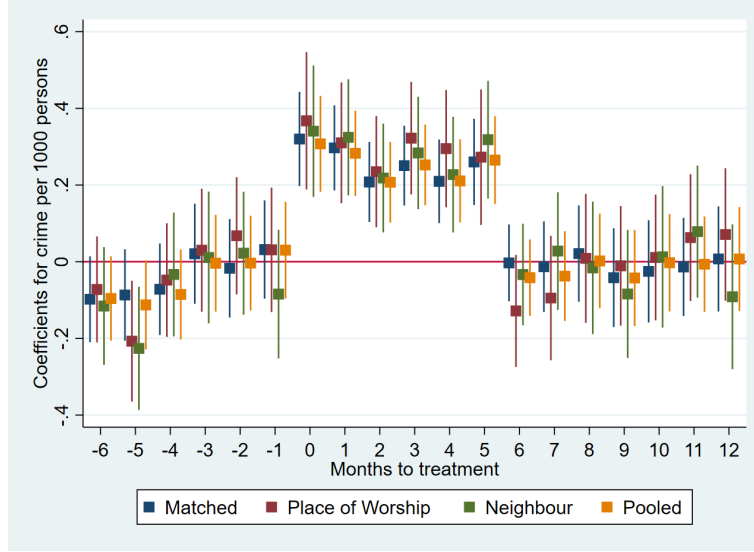
Notes: A plot of estimates for the average treatment effect on the treated (ATT) salience on violent crime split by whether the share of votes going to broadly right-wing or anti-immigrant parties or agendas was lower or higher than the median, as well as a composition-adjusted estimate based on the lower share constituencies. A 95% confidence interval is included around each estimate.

This suggests that voting behaviour and hence political beliefs may play an independent role in pushing individuals towards committing violent crimes. It has been suggested, for example, that exposure to hyperbolic and inflammatory language can increase the willingness of followers to express overly hostile views towards minorities and increase costs of assimilation (Bursztyń et al., 2020; Gould and Klor, 2016). In recent decades, this has been exacerbated by the spread of social media, sometimes resulting in the spread of hate crime (Müller and Schwarz, 2021, 2023).

4.4 Robustness

I now discuss the robustness of my baseline results. I start by checking whether these are sensitive to the use of alternative samples of control units. Figure 13 presents the event study estimates for the effect of terrorist attacks on violent crime based on four different samples: (i) 1:3 propensity score matched, (ii) Neighbouring areas within 500m of a treated area, (iii) Areas with non-mosque places of worship and (iv) All aforementioned control units pooled together. With the exception of the $t = -5$ pre-period estimate, the wider set of results are remarkably consistent across all four control unit samples implying that the internal validity of the estimates does not hinge on the success of the propensity score matching strategy.

Figure 13: Event Study Plot for Violent Crime

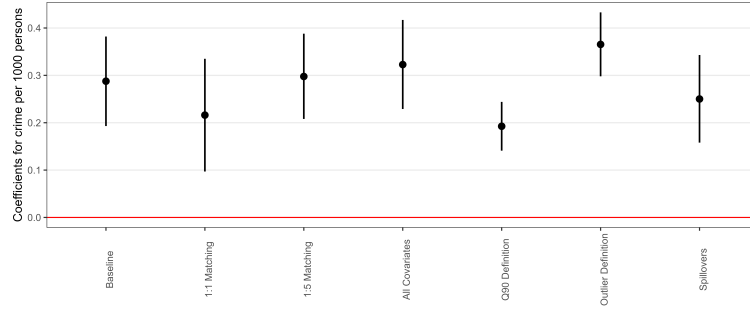


Notes: An event study plot for the effect of Islamic terrorist attacks on the rate of violent crime. Estimation is based on four different samples: (i) 1:3 propensity score matched, (ii) Neighbouring areas within 500m, (iii) Areas with non-mosque places of worship and (iv) All aforementioned units pooled together. A 95% confidence interval is included around each estimate.

Next, I consider whether the results are driven by my choice of matching ratio, covariate choice, treatment definition or by spatial spillover effects. Hence, I compare the baseline ATT estimates to several alternative specifications. To start with, I adduce estimates based on 1:1 and 1:5 propensity score matching ratios of treated to control units, as well as those using a 1:3 matching ratio but with an expanded set of covariates used in the matching process. Next, I use two alternative definitions of exposure. The first is whether a neighbourhood is within the 90th percentile of its local authority for the number of Muslim residents, while the second is whether this number is an outlier ($1.75 \times IQR$) for its local authority. Finally, I check whether the results are driven by spatial spillover effects. To do so, I modify the matching procedure and exclude all units within 1.5km of a treated unit from the pool of controls.

Figure 14 presents the entire sequence of ATT estimates for each of the robustness checks mentioned above. It is clear from a visual inspection that changes to the matching process by either altering the ratio of treated to control units, not using neighbouring areas as controls or the use of additional covariates do not significantly alter the magnitude of the results. When considering changes to the treatment definition, though the estimates vary because there is a change in the composition of treated and untreated units, the estimated effects are large and positive across all specifications. Indeed, the outlier definition yields the largest effects. However, in each case the estimates are not statistically different from those yielded by my preferred definition.

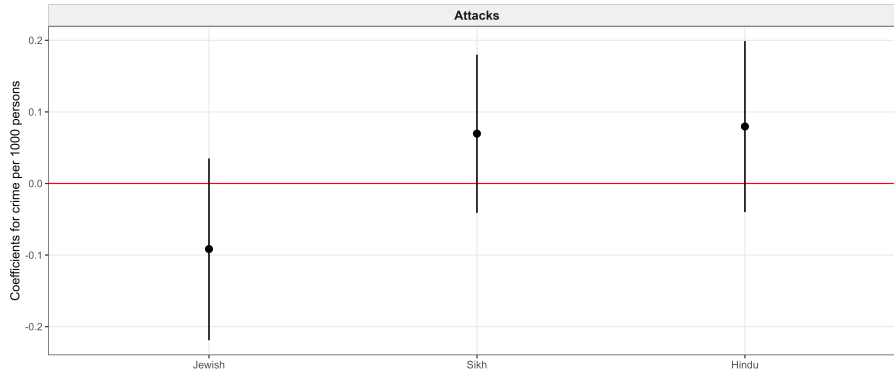
Figure 14: Robustness Checks



Notes: A plot of estimates for the average treatment effect on the treated (ATT) for the effect of Islamic terrorist attacks on violent crime. The baseline 1:3 propensity score matched approach where a treated unit is defined to be one close to a mosque and within 75th percentile of LAD is compared to six other specifications. Namely, four alternative matching approaches (1:1, 1:5, removal of units potentially affected by spillovers, and a specification with an expanded covariate set), and two definitions of exposure to salience shocks (share of Muslims being in the 90th percentile of the district, and share of Muslim residents being an outlier in the district). A 95% confidence interval is included around each estimate.

Finally, I conduct a falsification exercise which checks whether other minority groups - Jewish, Sikh or Hindu communities - are also affected by the shocks to Muslim salience. It is expected that results should be zero unless affected by spillover effects either spatially or through changes to who is targeted by typical violent hate crime perpetrators. The results are presented in Figure 15. The point estimates are not statistically significant at the 5% level, thus supporting the validity of my baseline results.

Figure 15: Falsification Test based on Placebo Groups



Notes: A plot of estimates for the average treatment effect on the treated (ATT) of terrorist attacks on violent crime. The 1:3 propensity score matched approach is used for three separate placebo groups (Jews, Sikhs and Hindus) where a neighbourhood is considered treated if the number of religiously affiliated residents is in the 90th percentile for the district. A 95% confidence interval is included around each estimate.

5 Conclusion

In this paper, I use neighbourhood-level crime data to investigate how terrorism-induced changes in Muslim salience affect local crime rates. Previous research has used more limited geographic variation, leveraging differences in crime trends across groups of po-

tential victims (Ivandic et al., 2024). Hence, a key contribution of this paper is the detailed exploration of treatment effect heterogeneity through the use of country-wide police data. Understanding which factors make violent backlash more or less likely is crucial from a policy perspective, particularly when promoting integration of religious minorities, and is central to designing policies that protect vulnerable communities and preserve social cohesion.

The main finding is that changes to Muslim salience cause a large and significant increase in violent crime around Muslim communities. Using a matched difference-in-difference approach, I find that in the six months following an Islamic terrorist attack, violent crime, in neighbourhoods with large and visible Muslim communities, rises by roughly 12-13% before returning to its pre-attack trend. The absence of pre-trends and the stability of the estimates across alternative control groups, matching specifications and treatment definitions supports a causal interpretation of these results. Newspaper coverage and Google search intensity produce smaller but significant effects. The newspaper effect rises monotonically with coverage intensity, which suggests that the volume of media attention, and not just the attack itself, shapes the size of the backlash. Placebo tests on Jewish, Sikh and Hindu communities yield no significant effects, while the rise in serious A&E admissions suggests the results are not driven by changes in reporting or policing alone. In addition, the intensification of the Israeli-Palestinian conflict in October 2023 was followed by a significant rise in violent crime around Jewish communities, highlighting that the link between salience and violence is not specific to Muslim communities, and can operate in other groups in society.

Heterogeneity analysis reveals that neighbourhood and constituency-level characteristics explain little of the variation in backlash challenging economic conflict accounts of anti-minority violence (Bray et al., 2024). Nonetheless, two neighbourhood characteristics matter: religious fractionalisation, which amplifies the response, while a high share of young residents dampens it. In addition, voting behaviour, reflecting political attitudes and preferences, also appears to play a role with violent crime responding more strongly in areas with wider support for anti-immigrant parties. A re-weighting technique shows that this gap is not accounted for by observable differences in social, economic or demographic composition, suggesting that the political beliefs act as an independent channel between salience and crime (Clifton-Sprigg et al., 2020).

These results have several policy implications. External threats, perceived or otherwise, can create strong tension between different ethnic and religious communities even in formerly well-integrated countries such as the UK. Hence, interventions aimed at community cohesion may be most effective if concentrated in the months after an attack, in religiously diverse areas and in areas where anti-immigrant sentiment is strong. The role of media coverage highlights a difficult trade-off between press freedom and social harmony, and points to the potential value of responsible reporting guidelines. The spillovers to other minorities across neighbourhoods suggest that protective policing measures should extend beyond the directly targeted group.

References

- Abadie, Alberto**, “Semiparametric Difference-in-Differences Estimators,” *The Review of Economic Studies*, January 2005, 72 (1), 1–19.
- **and Jann Spiess**, “Robust Post-Matching Inference,” *Journal of the American Statistical Association*, April 2022, 117 (538), 983–995.
- Adão, Rodrigo, Michal Kolesár, and Eduardo Morales**, “Shift-Share Designs: Theory and Inference*,” *The Quarterly Journal of Economics*, November 2019, 134 (4), 1949–2010.
- Alabrese, Eleonora, Sascha O. Becker, Thiemo Fetzer, and Dennis Novy**, “Who voted for Brexit? Individual and regional data combined,” *European Journal of Political Economy*, January 2019, 56, 132–150.
- Alba, R., R. G. Rumbaut, and K. Marotz**, “A Distorted Nation: Perceptions of Racial/Ethnic Group Sizes and Attitudes Toward Immigrants and Other Minorities,” *Social Forces*, December 2005, 84 (2), 901–919.
- Alesina, Alberto and Eliana La Ferrara**, “Who trusts others?,” *Journal of Public Economics*, August 2002, 85 (2), 207–234.
- **and –**, “Ethnic Diversity and Economic Performance,” *Journal of Economic Literature*, August 2005, 43 (3), 762–800.
- **and Marco Tabellini**, “The Political Effects of Immigration: Culture or Economics?,” *Journal of Economic Literature*, March 2024, 62 (1), 5–46.
- **, Armando Miano, and Stefanie Stantcheva**, “Immigration and Redistribution,” *The Review of Economic Studies*, January 2023, 90 (1), 1–39.
- Alrababa’H, Ala’, William Marble, Salma Mousa, and Alexandra A. Siegel**, “Can Exposure to Celebrities Reduce Prejudice? The Effect of Mohamed Salah on Islamophobic Behaviors and Attitudes,” *American Political Science Review*, November 2021, 115 (4), 1111–1128.
- Anderson, D. Mark, Benjamin Crost, and Daniel I. Rees**, “Do economic downturns fuel racial animus?,” *Journal of Economic Behavior & Organization*, July 2020, 175, 9–18.
- Beatty, Christina and Steve Fothergill**, “Hitting the poorest places hardest: The local and regional impact of welfare reform,” Technical Report, Sheffield Hallam University April 2013.
- Becker, Sascha O and Thiemo Fetzer**, “Does Migration Cause Extreme Voting?,” *CAGE Working Paper 306*, 2017.
- **, – , and Dennis Novy**, “Who voted for Brexit? A comprehensive district-level analysis,” *Economic Policy*, October 2017, 32 (92), 601–650.
- **, – , and –**, “Erratum to: Who voted for Brexit? A comprehensive district-level analysis,” *Economic Policy*, January 2018, 33 (93), 179–180.
- Bonomi, Giampaolo, Nicola Gennaioli, and Guido Tabellini**, “Identity, Beliefs, and Political Conflict,” *The Quarterly Journal of Economics*, October 2021, 136 (4), 2371–2411.

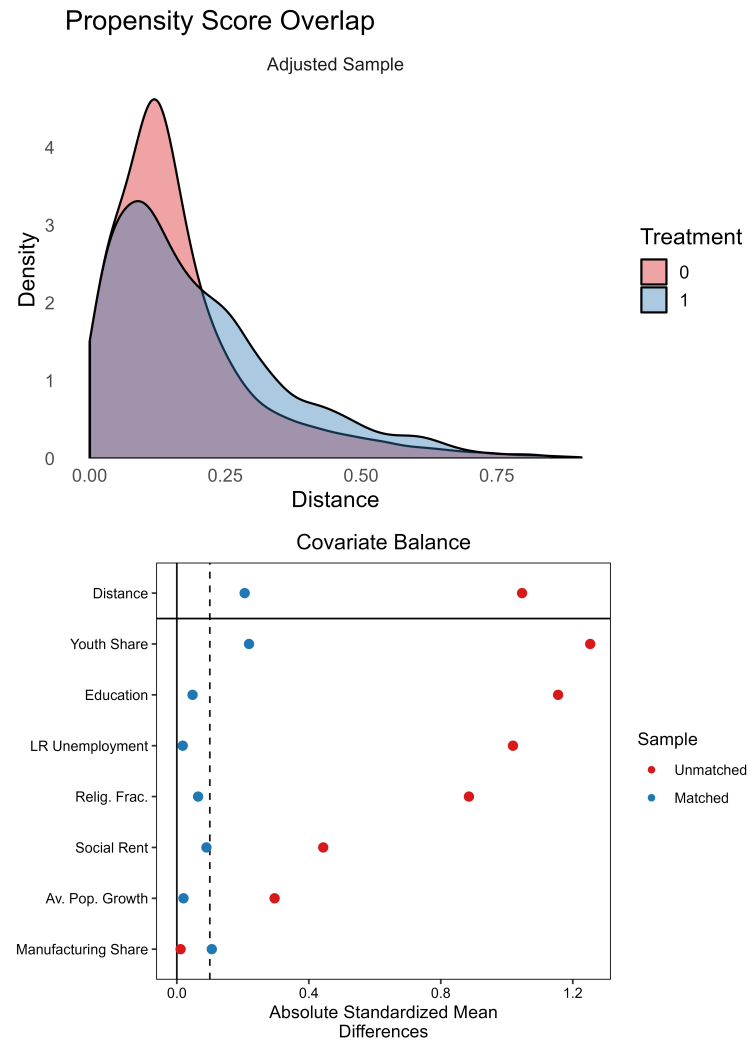
- Bordalo, Pedro, Nicola Gennaioli, and Andrei Shleifer**, “Salience and Consumer Choice,” *Journal of Political Economy*, October 2013, 121 (5), 803–843.
- Bray, Kerry, Nils Braakmann, and John Wildman**, “Austerity, welfare cuts and hate crime: Evidence from the UK’s age of austerity,” *Journal of Urban Economics*, May 2024, 141, 103439.
- Bursztyn, Leonardo, Georgy Egorov, and Stefano Fiorin**, “From Extreme to Mainstream: The Erosion of Social Norms,” *American Economic Review*, November 2020, 110 (11), 3522–3548.
- , – , **Ruben Enikolopov, and Maria Petrova**, “Social Media and Xenophobia: Evidence from Russia,” Technical Report w26567, National Bureau of Economic Research, Cambridge, MA December 2019.
- Böhmelt, Tobias, Vincenzo Bove, and Enzo Nussio**, “Can Terrorism Abroad Influence Migration Attitudes at Home?,” *American Journal of Political Science*, July 2020, 64 (3), 437–451.
- Callaway, Brantly, Andrew Goodman-Bacon, and Pedro H. Sant’Anna**, “Difference-in-differences with a Continuous Treatment,” Technical Report w32117, National Bureau of Economic Research, Cambridge, MA February 2024.
- Clifton-Sprigg, Joanna, Jonathan James, Suncica Vujic, and Joel Carr**, “Love Thy Neighbour? Brexit and Hate Crime,” *SSRN Electronic Journal*, 2020.
- Colussi, Tommaso, Ingo E. Isphording, and Nico Pestel**, “Minority Salience and Political Extremism,” *American Economic Journal: Applied Economics*, July 2021, 13 (3), 237–271.
- Deloughery, Kathleen, Ryan D. King, and Victor Asal**, “Close Cousins or Distant Relatives? The Relationship Between Terrorism and Hate Crime,” *Crime & Delinquency*, September 2012, 58 (5), 663–688.
- DeMarzo, P. M., D. Vayanos, and J. Zwiebel**, “Persuasion Bias, Social Influence, and Unidimensional Opinions,” *The Quarterly Journal of Economics*, August 2003, 118 (3), 909–968.
- DiNardo, John, Nicole M. Fortin, and Thomas Lemieux**, “Labor Market Institutions and the Distribution of Wages, 1973-1992: A Semiparametric Approach,” *Econometrica*, September 1996, 64 (5), 1001.
- Dustmann, Christian, Francesca Fabbri, and Ian Preston**, “Racial Harassment, Ethnic Concentration, and Economic Conditions*,” *The Scandinavian Journal of Economics*, September 2011, 113 (3), 689–711.
- Facchetti, Elisa**, “Police Infrastructure, Police Performance, and Crime: Evidence from Austerity Cuts,” *SSRN Electronic Journal*, 2025.
- Falk, Armin, Andreas Kuhn, and Josef Zweimüller**, “Unemployment and Right-wing Extremist Crime*,” *The Scandinavian Journal of Economics*, June 2011, 113 (2), 260–285.
- Fetzer, Thiemo**, “Did Austerity Cause Brexit?,” *American Economic Review*, November 2019, 109 (11), 3849–3886.
- Giulietti, Corrado and Brendon McConnell**, “Kicking You When You’re Already Down: The Multipronged Impact of Austerity on Crime,” 2020. Version Number: 4.

- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift**, “Bartik Instruments: What, When, Why, and How,” *American Economic Review*, August 2020, 110 (8), 2586–2624.
- Gould, Eric D. and Esteban F. Klor**, “The Long-run Effect of 9/11: Terrorism, Backlash, and the Assimilation of Muslim Immigrants in the West,” *The Economic Journal*, November 2016, 126 (597), 2064–2114.
- Guriev, Sergei and Elias Papaioannou**, “The Political Economy of Populism,” *Journal of Economic Literature*, September 2022, 60 (3), 753–832.
- Hanes, Emma and Stephen Machin**, “Hate Crime in the Wake of Terror Attacks: Evidence From 7/7 and 9/11,” *Journal of Contemporary Criminal Justice*, August 2014, 30 (3), 247–267.
- Illing, Hannah, Johannes Schmieder, and Simon Trenkle**, “The Gender Gap in Earnings Losses After Job Displacement,” *Journal of the European Economic Association*, March 2024, p. jvae019.
- Ivandic, Ria, Tom Kirchmaier, and Stephen Machin**, “International Terror Attacks and Local Out-Group Hate Crimes,” *The Journal of Law and Economics*, August 2024, 67 (3), 589–610.
- Krueger, Alan B. and Jorn-Steffen Pischke**, “A Statistical Analysis of Crime against Foreigners in Unified Germany,” *The Journal of Human Resources*, 1997, 32 (1), 182.
- Liberini, Federica, Andrew J. Oswald, Eugenio Proto, and Michela Redoano**, “Was Brexit triggered by the old and unhappy? Or by financial feelings?,” *Journal of Economic Behavior & Organization*, May 2019, 161, 287–302.
- Müller, Karsten and Carlo Schwarz**, “Fanning the Flames of Hate: Social Media and Hate Crime,” *Journal of the European Economic Association*, August 2021, 19 (4), 2131–2167.
- and —, “From Hashtag to Hate Crime: Twitter and Antiminority Sentiment,” *American Economic Journal: Applied Economics*, July 2023, 15 (3), 270–312.
- Rushin, Stephen and Griffin Sims Edwards**, “The Effect of President Trump’s Election on Hate Crimes,” *SSRN Electronic Journal*, 2018.
- Schneider-Strawczynski, Sarah and Jérôme Valette**, “Media Coverage of Immigration and the Polarization of Attitudes,” *American Economic Journal: Applied Economics*, January 2025, 17 (1), 337–368.
- Shayo, Moses**, “Social Identity and Economic Policy,” *Annual Review of Economics*, August 2020, 12 (1), 355–389.
- Swahn, M H, R R Mahendra, L J Paulozzi, R L Winston, G A Shelley, J Taliano, L Frazier, and J R Saul**, “Violent attacks on Middle Easterners in the United States during the month following the September 11, 2001 terrorist attacks: Figure 1,” *Injury Prevention*, June 2003, 9 (2), 187–189.
- Villa, Carmen**, “The effects of youth clubs on education and crime,” Technical Report, The IFS November 2024.
- Walters, Mark and Alex Krasodonski-Jones**, “Patterns of Hate Crime: Who, What, When and Where?,” 2018.

Appendix

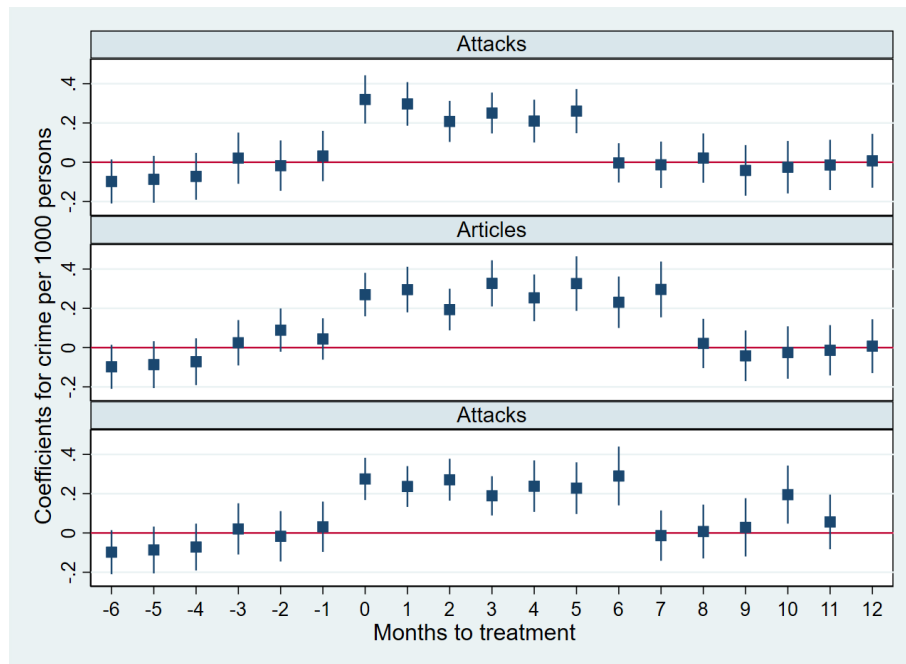
Figures

Figure A.1: Overlap and Balance Plots



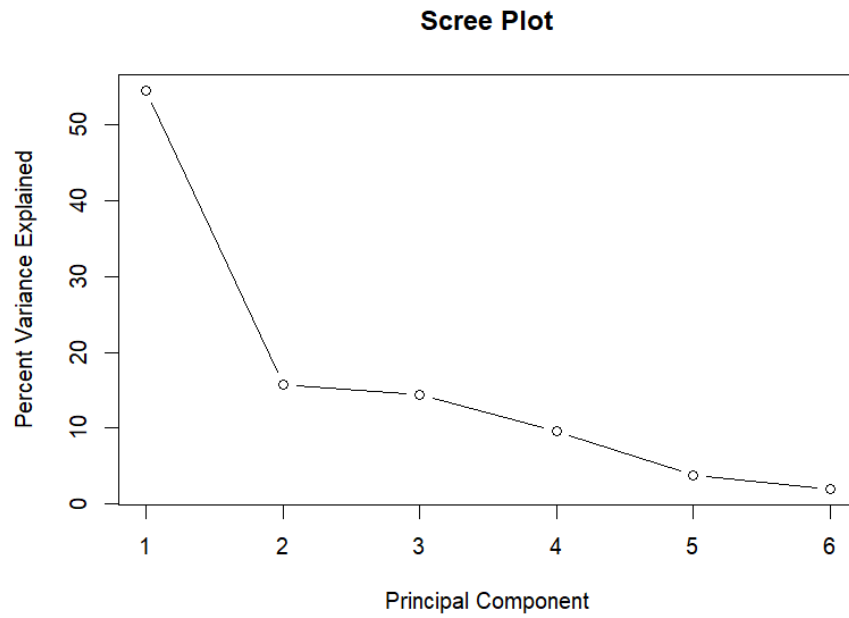
Notes: Plots illustrating propensity score overlap between treated and untreated units, as well as a plot showing covariate balance before (red) and after (blue) matching.

Figure A.2: Event Study Plots for Violent Crime



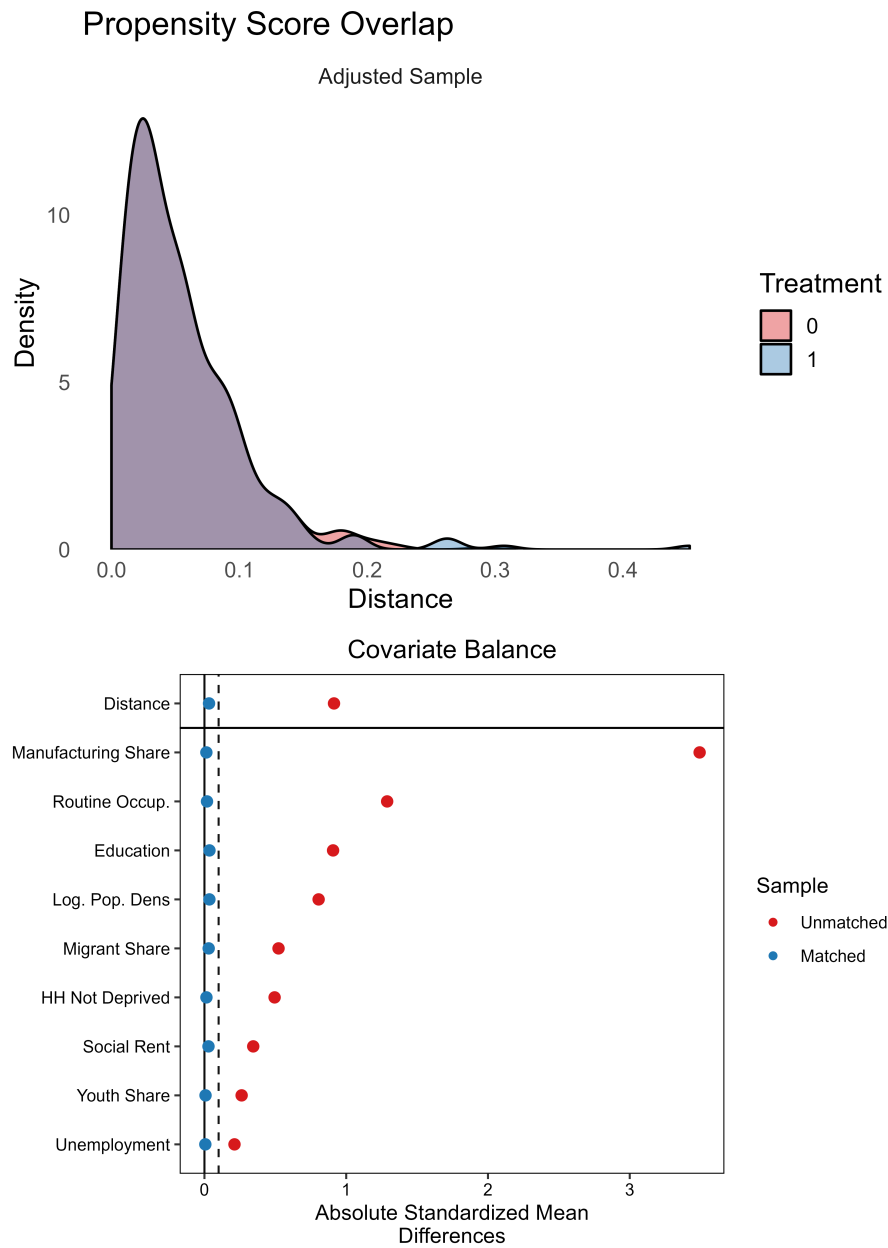
Notes: An event study plot for the effect of changes to Muslim salience on the rate of violent crime for all three measures of salience: terrorist attack dummy, newspaper article dummy, and Google search dummy. Estimation is based on a 1:3 propensity score matched sample. A 95% confidence interval is included around each estimate.

Figure A.3: Scree Plot



Notes: A scree plot for the principal components analysis used to reduce the number of dimensions of a set of socio-economic variables when analysing the role of voting behaviour in the salience-crime relationship.

Figure A.4: Overlap and Balance Plots



Notes: Plots illustrating propensity score overlap between treated and untreated units, as well as a plot showing covariate balance before (red) and after (blue) matching.

Tables

Table A.1: Control Variable Descriptions

Variable	Explanation
Population Density	The number of residents of a neighbourhood per square kilometre.
Education	The share of individuals who have achieved at least an GCSE of education.
Youth Share	The share of residents aged 0–24.
Long-Term Unemployment	The share of residents that have not been able to find a job in 5+ years
Religious Fractionalization	Derived based on the formula in Alesina and La Ferrara (2002) that measures the probability that any two individuals from a neighbourhood will have different religions.
Foreign Share	Proportion of residents that were born outside the UK.
Delta Foreign Share	Change in the proportion of residents that were born outside the UK from 2001 to 2011.
Africa & ME Share	Proportion of residents that were born in Africa or the Middle East.
Manufacturing Share	Proportion of residents employed in the manufacturing industry.
Full-time Share	Proportion of residents in full-time employment.
Social Rent Share	Share of individuals living in socially-rented accommodation.
Austerity Loss	Estimated per person loss (GBP) from austerity cuts calculated by Beatty and Fothergill (2013). Note that for some local authorities, these values are not available.

Table A.2: Heterogeneity Analysis - Constituency-level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Treat	0.139 (0.122)	0.320*** (0.047)	0.286*** (0.099)	0.106 (0.102)	0.296** (0.136)	0.345* (0.181)	0.293*** (0.135)	0.256* (0.137)	0.233 (0.192)
Treat x Pop Dens.	0.187 (0.133)								
Treat x Educ.		-0.147 (0.166)							
Treat x Soc. Rent			-0.008 (0.113)						
Treat x Unemp.				0.199* (0.116)					
Treat x Prop. Youth					0.001 (0.146)				
Treat x Relig. Frac						-0.038 (0.187)			
Treat x Foreign							0.048 (0.145)		
Treat x Delta Foreign								0.073 (0.147)	
Treat x ME & Africa Share									0.128 (0.199)
P	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
N	433326	433326	433326	433326	433326	433326	433326	433326	433326

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the constituency-level heterogeneity analysis estimates for the average treatment effect on the treated of changes to Muslim salience on the monthly crime rate at the constituency-level. Each model uses time, time by characteristics and neighbourhood fixed effects in a triple difference-in-difference-style design. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the matched sub-group level.

Table A.3: Robustness Check - Corroborative Approach - Alternative Treatment

	(1) Mosq.	(2) Mosq.	(3) Mosq.	(4) Q90	(5) Q90	(6) Q90	(7) Out.	(8) Out.	(9) Out.
Attack x Q75 Mosq.	0.177*** (0.022)								
Articles x Mosq.		0.048*** (0.018)							
Google x Mosq.			0.013 (0.017)						
Attack x Q90				0.065*** (0.014)					
Articles x Q90					0.055*** (0.012)				
Google x Q90						-0.006 (0.011)			
Attack x Out.							0.076*** (0.018)		
Articles x Out.								0.077*** (0.015)	
Google x Out.									0.015 (0.014)
N	2675711	2675711	2675711	2675711	2675711	2675711	2675711	2675711	2675711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. Three measure of exposure are used in these regressions: whether the size of the local Muslim community in the local authority is in the 75th percentile, and there is a Mosque in the vicinity (within 500m) whether the size of the local Muslim community in the local authority is in the 90th percentile, and whether the size of the Muslim community is an outlier in the local authority. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of violent crime defined as the number of Police reported offences per 1000 residents per month respectively. Three measures of salience are used: terrorist attacks, newspaper articles and the Google Trends search index. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

Table A.4: Falsification Test - Corroborative Approach - Placebo Groups

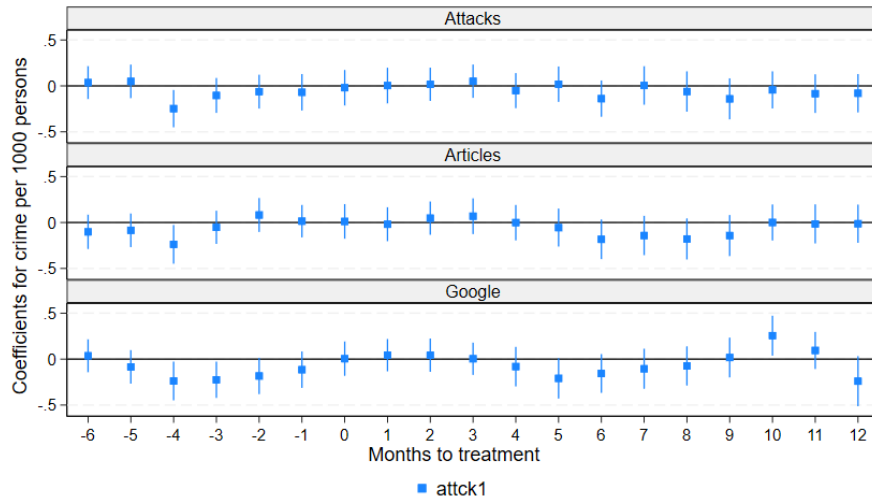
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Attack x Jew	-0.016 (0.011)								
Articles x Jew		-0.020** (0.009)							
Google x Jew			-0.005 (0.008)						
Attack x Sikh				0.022** (0.010)					
Articles x Sikh					-0.013 (0.009)				
Google x Sikh						-0.011 (0.008)			
Attack x Hindu							0.004 (0.013)		
Articles x Hindu								0.003 (0.011)	
Google x Hindu									-0.011 (0.010)
N	2675711	2675711	2675711	2675711	2675711	2675711	2675711	2675711	2675711

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the estimation results illustrating the average treatment effect of changes to muslim salience on the monthly crime rate at the LSOA-level. The measure of exposure used in these regression is whether the size of the local Jewish (models 1-3), Sikh (models 4-6) or Hindu (models 7-9) community in the local authority is in the 90th percentile. Each model uses month and neighbourhood fixed effects, and fits a neighbourhood-specific quadratic trend to the crime rate. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. Standard errors are reported in parentheses and are clustered at the neighbourhood-level.

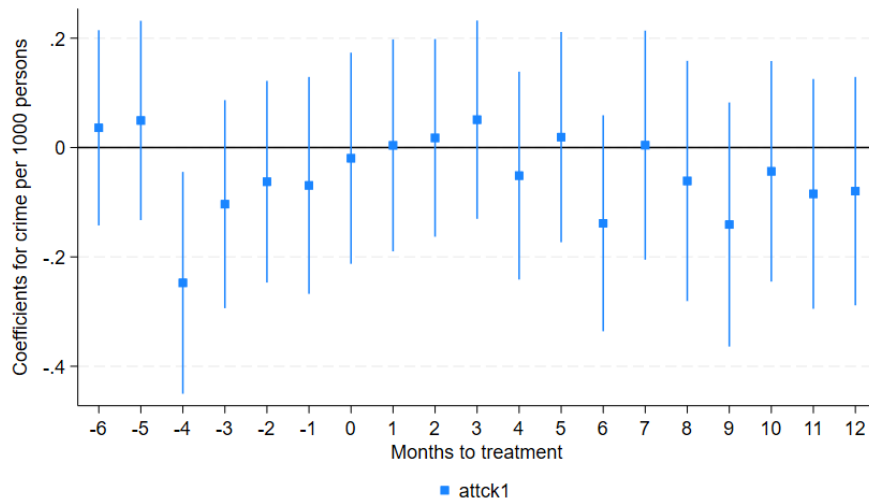
Online Appendix

Figure B.1: Event Study Plot for ASB



Notes: An event study plot for the effect of changes to Muslim salience on the rate of anti-social behaviour for all three measures of salience: terrorist attack dummy, newspaper article dummy, and Google search dummy. Estimation is based on four different samples: (i) 1:3 propensity score matched, (ii) Neighbouring areas within 500m, (iii) Areas with non-mosque places of worship and (iv) All aforementioned units pooled together. A 95% confidence interval is included around each estimate.

Figure B.2: Event Study Plot for ASB



Notes: An event study plot for the effect of changes to Muslim salience measured using the occurrence of a terrorist attack on the rate of anti-social behaviour (ASB). A 95% confidence interval is included around each estimate.

Table B.1: Saliency and Anti-Social Behaviour (ASB)

	(1)	(2)	(3)
Attack x Treat	0.042 (0.048)		
Articles x Treat		0.043 (0.036)	
Google x Treat			0.032 (0.034)
p	0.376	0.236	0.343
N	594642	594642	594642

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Notes: This table reports the baseline difference-in-differences estimates for the average treatment effect on the treated of changes to Muslim saliency on the monthly crime rate at the LSOA-level. Each model uses time and neighbourhood fixed effects. The models report the effects on the rate of violent crime defined as the number of police reported offences per 1000 residents per month respectively. Three measures of saliency are used: terrorist attacks, newspaper articles and the Google Trends search index. For each measure of saliency, four models are run based on different samples as the control group: matched, place of worship, neighbourhood and pooled. Standard errors are reported in parentheses and are clustered at the individual-level or matched sub-group level.